▶ RECIPE = follow the steps | E example | F citable fact | D definition | ! trap

The task says...	Jump to
"Berechnen Sie den Grenzwert" ($x \to 0$, $0/0$)	**5.6** Taylor-plug method (faster + safer than l'Hospital)
"Grenzwert einer Folge"	**2.2** limit recipe, **2.3** recursive $x_{n+1} = f(x_n)$
"Konvergiert die Reihe?"	**3.2** decision tree
"Konvergenzradius / für welche x"	**3.3** radius recipe
"Bestimmen Sie $\int f\,dx$"	**6.1** integral decision tree
"allgemeine Lösung der DGL"	**7.1** ODE decision tree
"Kurvendiskussion / Extrema"	**5.4** full recipe
"Häufungspunkt / limsup"	**2.4** subsequence split, **2.5** Cauchy
"konvergiert (f_n) gleichmässig?"	**4.7** sup-norm recipe
"punktweiser Grenzwert"	**4.6** fix x, let $n \to \infty$
"gleichmässig stetig?"	**4.5** recipe + case distinction
"Riemann-integrierbar?"	**6.7** criteria list
"uneigentliches Integral konvergent?"	**6.8** p-test + comparison
"Fixpunkt / eindeutige Nullstelle"	**4.4** IVT + strict monotonicity
"Beweisen Sie ... (Vorlesung)"	**8** proof skeletons
"differenzierbar in x_0?"	**5.1** + Landau o/O box

1 Toolbox: numbers, sup/inf, complex, algebra

1.1 sup / inf / max / min (Supremum, Infimum)

DEF: the four words

$M \subseteq \mathbb{R}$ nonempty. **upper bound** s: $x \le s\ \forall x \in M$. $\sup M$ = smallest upper bound. $\max M = \sup M$ *if* $\sup M \in M$. Same mirrored for inf / min. **Completeness axiom:** every nonempty M bounded above has a $\sup M \in \mathbb{R}$ (this is THE property $\mathbb{Q}$ lacks).

▶ RECIPE: find sup / inf and prove it

1 Guess the value s (largest/smallest value the formula can reach).
2 **Upper bound:** show $x \le s$ for all $x \in M$ (algebra).
3 **Least:** $\forall \varepsilon > 0$ find $x \in M$ with $x > s - \varepsilon$ (usually pick $n > 1/\varepsilon$).
4 $s \in M$? yes $\Rightarrow$ max = s; no $\Rightarrow$ max does not exist.

TRAP: sup/inf statements in multiple choice

- sup always *exists in* $\mathbb{R} \cup \{\pm\infty\}$; as a *real number* only if the set is bounded. Unbounded set $\Rightarrow$ "sup $= +\infty$", i.e. no real sup.
- $\min M$ is **not** automatically an accumulation point (it can be attained once, e.g. $x_0 = 0$, $x_n = 1 + 1/n$).
- $\inf M = \sup M \Rightarrow$ set has exactly one element $\Rightarrow$ sequence is constant $\Rightarrow$ converges. **True.**
- monotone *and bounded* $\Rightarrow$ converges to sup (incr.) / inf (decr.). Monotone alone $\Rightarrow$ may run to $\pm\infty$.

1.2 Induction (vollständige Induktion)

▶ RECIPE: proof by induction

1 **Base:** check $n = n_0$ explicitly.
2 **Hypothesis:** assume $A(n)$ true for a fixed $n \ge n_0$.
3 **Step:** write the $n+1$ statement, isolate the n part, replace it using the hypothesis, then estimate/simplify to the target.

Standard helpers: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$, Bernoulli $(1+x)^n \ge 1 + nx$ $(x \ge -1)$.

1.3 Inequality toolkit (Abschätzungen)

FACT: the five you actually use

- **Triangle:** $|a+b| \le |a| + |b|$; **reverse:** $\big||a| - |b|\big| \le |a - b|$.
- **Bernoulli:** $(1+x)^n \ge 1 + nx$ for $x \ge -1$.
- **AM–GM:** $\sqrt{ab} \le \frac{a+b}{2}$ $(a, b \ge 0)$.
- **Young:** $ab \le \frac{a^2}{2} + \frac{b^2}{2}$, more generally $ab \le \frac{\varepsilon a^2}{2} + \frac{b^2}{2\varepsilon}$.
- **Cauchy–Schwarz:** $|\langle x, y\rangle| \le \|x\|\,\|y\|$.

Always usable: $|\sin t| \le 1$, $|\sin t| \le |t|$, $|\cos t| \le 1$, $e^t \ge 1 + t$, $\ln t \le t - 1$.

TRAP: how to estimate without panic

Goal is always $|\text{expr}| \le (\text{something} \to 0)$.

- Numerator: make it *bigger* (drop $-$ terms, bound sin / cos by 1).
- Denominator: make it *smaller* (drop + terms) — never the other way round.
- Roots: multiply by the conjugate $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A}+\sqrt{B}}$.

1.4 Complex numbers $\mathbb{C}$

DEF: forms and conversion

$z = a + ib = re^{i\varphi} = r(\cos\varphi + i\sin\varphi)$, $\bar{z} = a - ib$, $|z| = r = \sqrt{a^2+b^2}$, $z\bar{z} = |z|^2$. **Cartesian→polar:** $r = \sqrt{a^2+b^2}$, $\varphi = \operatorname{atan2}(b, a)$ i.e. $\arctan(b/a)$ *plus* π *if* $a < 0$. ! quadrant check! **Polar→Cartesian:** $a = r\cos\varphi$, $b = r\sin\varphi$. **Division:** multiply by $\frac{\bar{w}}{\bar{w}}$: $\frac{z}{w} = \frac{z\bar{w}}{|w|^2}$. **Euler:** $e^{i\varphi} = \cos\varphi + i\sin\varphi$; $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

▶ RECIPE: solve $z^n = w$ (n-th roots)

1 Write $w = \rho e^{i\psi}$ (get ρ, ψ as above).
2 $z_k = \rho^{1/n} e^{i(\psi + 2\pi k)/n}$, $k = 0, 1, \dots, n-1$.
3 Exactly n roots, all on the circle of radius $\rho^{1/n}$, spaced by $2\pi/n$.

$\sqrt[n]{1}$: $z_k = e^{2\pi i k/n}$. Sum of all n-th roots of unity $= 0$ $(n \ge 2)$.

FACT: polynomials — what is guaranteed

p of degree n with *any* complex coefficients:

- has **exactly** n **roots in** $\mathbb{C}$ **counted with multiplicity** (Fundamental Thm. of Algebra) — not necessarily distinct.
- ! Conjugate pairing $\bar{z}$ root whenever z is — **only if all coefficients are real**.
- Odd degree *and real* coefficients $\Rightarrow$ at least one real root (IVT). Degree 5 with complex coefficients: *no* real root guaranteed.

▶ RECIPE: quadratic with complex data

$az^2 + bz + c = 0 \Rightarrow z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$; for complex $D = b^2 - 4ac$ compute $\sqrt{D}$ by the n-th-root recipe with $n = 2$. Real coefficients, $D < 0$: $z = \frac{-b}{2a} \pm i\frac{\sqrt{|D|}}{2a}$.

2 Sequences (a_n)

2.1 Definitions you must be able to write

DEF: convergence / divergence / boundedness

$a_n \to a :\iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \ge N: |a_n - a| < \varepsilon$.
$a_n \to \infty :\iff \forall K > 0\ \exists N\ \forall n \ge N: a_n > K$.
bounded: $\exists C: |a_n| \le C\ \forall n$. **monotone incr.:** $a_{n+1} \ge a_n\ \forall n$.
Chain of implications: convergent $\Rightarrow$ Cauchy $\Rightarrow$ bounded $\Rightarrow$ has a convergent subsequence (Bolzano–Weierstrass). None of the arrows reverses except Cauchy$\Leftrightarrow$convergent in $\mathbb{R}$.

2.2 Limit of an explicitly given a_n

▶ RECIPE: compute $\lim a_n$

1 **Rational in** n: divide num. & den. by the highest power of n in the *denominator*. Degree top < bottom $\Rightarrow$ 0; = $\Rightarrow$ ratio of leading coeff.; > $\Rightarrow \pm\infty$.
2 **Roots / $\infty - \infty$:** multiply by conjugate $\frac{\sqrt{A}+\sqrt{B}}{\sqrt{A}+\sqrt{B}}$.
3 **n-th roots / powers:** write $a_n = e^{\ln a_n}$ and take the limit in the exponent (continuity of exp).
4 $(1 + \frac{c}{n})^{dn}$ **shape:** $\to e^{cd}$.
5 **Factorials / n^n:** use growth order $\ln n \ll n^\alpha \ll b^n \ll n! \ll n^n$ $(\alpha > 0, b > 1)$.
6 **sin / cos or $(-1)^n$ times a null sequence:** squeeze $-|b_n| \le a_n \le |b_n|$.
7 **Else:** treat n as real x, use l'Hospital / Taylor (§5.5–5.6), transfer back.

FACT: tools with names (cite them!)

Limit laws: sums, products, quotients (den. $\ne 0$) — only valid if *both* limits exist finitely. **Squeeze** *(Sandwich)*: $b_n \le a_n \le c_n$, $b_n, c_n \to L \Rightarrow a_n \to L$. **Monotone convergence** *(Monotoniekriterium)*: monotone + bounded $\Rightarrow$ convergent, limit = sup resp. inf of the set of terms. **Continuity transfer:** $a_n \to a$, f continuous at $a \Rightarrow f(a_n) \to f(a)$.

EXAMPLE: the seven limits that keep coming back

- $\frac{3n^2 - n}{2n^2 + 5} = \frac{3 - \frac{1}{n}}{2 + \frac{5}{n^2}} \to \frac{3}{2}$.
- $\sqrt{n^2+n} - n = \frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}$.
- $\sqrt[n]{n} = e^{\frac{\ln n}{n}} \to e^0 = 1$; likewise $\sqrt[n]{a} \to 1$ $(a > 0)$, $\sqrt[n]{2^n + 3^n} \to 3$ (pull out the largest).
- $\left(1 + \frac{3}{n}\right)^{2n} = \left[\left(1 + \frac{3}{n}\right)^n\right]^2 \to (e^3)^2 = e^6$.
- $\frac{n!}{n^n} \le \frac{1}{n} \to 0$ (each of the other factors is ≤ 1).
- $\frac{2^n}{n!} \to 0$: for $n \ge 4$ the ratio $\frac{a_{n+1}}{a_n} = \frac{2}{n+1} \le \frac{1}{2} \Rightarrow$ geometric decay.
- $a_n = \frac{(-1)^n}{n}$: $|a_n| \le \frac{1}{n} \to 0 \Rightarrow a_n \to 0$ (squeeze; alternating alone means nothing).

▶ RECIPE: prove $a_n \to a$ by ε–N (when asked for a proof)

1 Compute $|a_n - a|$ and simplify to one fraction.
2 Over-estimate to a clean $\frac{C}{n^k}$ (numerator bigger, denominator smaller).
3 Solve $\frac{C}{n^k} < \varepsilon$ for $n \Rightarrow$ choose $N := \lceil (C/\varepsilon)^{1/k} \rceil + 1$.
4 Write: "Let $\varepsilon > 0$, choose N as above, then $\forall n \ge N$: $|a_n - a| \le \frac{C}{n^k} < \varepsilon$. $\square$"

2.3 Recursive sequences $x_{n+1} = f(x_n)$

▶ RECIPE: full treatment (this is a standard 4–9 point task)

1 **Candidate limit:** solve the fixed point equation $L = f(L)$. Keep all solutions.
2 **Bounded:** prove by induction that x_n stays in an interval I containing x_0 and L (show $f(I) \subseteq I$).
3 **Monotone:** compare x_1 with x_0; then induction: if f is increasing, the sign of $x_{n+1} - x_n$ never changes. Sign of $f(x) - x$ on I tells you the direction.
4 **Conclude:** monotone + bounded $\Rightarrow$ converges (Monotone Convergence). Limit L satisfies $L = f(L)$ because $x_{n+1} = f(x_n)$, f continuous, and $\lim x_{n+1} = \lim x_n$. Pick the root of $L = f(L)$ lying in I.

Alternative (contraction), often shorter: if $|f'(x)| \le q < 1$ on I then by MVT $|x_{n+1} - L| = |f(x_n) - f(L)| \le q|x_n - L| \Rightarrow |x_n - L| \le q^n|x_0 - L| \to 0$. Converges to the *unique* fixed point for *every* $x_0 \in I$.

EXAMPLE: $x_{n+1} = \frac{\sin x_n}{1 + x_n^2}$, $x_0 \in (0, 1)$

$f(x) = \frac{\sin x}{1+x^2}$. **1)** Fixed point: $x = 0$ works. For $x \ne 0$: $|\sin x| < |x|$ and $1 + x^2 > 1 \Rightarrow |f(x)| < |x| \Rightarrow$ no other fixed point. **2)** $x_0 \in (0, 1) \Rightarrow \sin x_0 > 0 \Rightarrow$ all $x_n > 0$; and $x_{n+1} = |f(x_n)| < x_n \Rightarrow$ strictly decreasing, bounded below by 0. **3)** Monotone + bounded $\Rightarrow$ converges to some $L \ge 0$ with $L = f(L) \Rightarrow L = 0$.
! Standard sub-question "show $|f(x)| < |x|$ for $x \ne 0$": use $|\sin x| < |x|$ for $x \ne 0$ (strict!) and $1 + x^2 \ge 1$.

2.4 Accumulation points, limsup, liminf

DEF: accumulation point (Häufungspunkt)

h is an accumulation point of (a_n) $:\iff$ every ε-ball around h contains a_n for *infinitely many* n $\iff$ some subsequence converges to h. $\limsup a_n =$ largest accumulation point $= \lim_{n\to\infty} \sup_{k\ge n} a_k$; $\liminf =$ smallest.

FACT: what you may conclude

- a_n converges $\iff$ bounded and $\liminf = \limsup$ $\iff$ exactly one accumulation point *and* bounded.
- **Bolzano–Weierstrass:** bounded $\Rightarrow$ at least one accumulation point.
- **Monotone sequence:** either bounded ($\Rightarrow$ converges, exactly 1 acc. point) or unbounded ($\Rightarrow \to \pm\infty$, **no** acc. point). So: "a monotone sequence *may have no* accumulation point" is the true statement; it can never have two.
- $a_n \le b_n\ \forall n \Rightarrow \lim a_n \le \lim b_n$ (**!** strict < is *not* preserved).

▶ RECIPE: find all accumulation points

1 Split by the periodic part: $(-1)^n \Rightarrow$ two subsequences $n = 2k$, $n = 2k+1$; $\cos(n\pi/3) \Rightarrow$ period 6; etc.
2 Compute the limit of each subsequence. That set = set of accumulation points.
3 $\limsup = \max$, $\liminf = \min$ of that set.

Ex.: $a_n = (-1)^n \frac{n}{n+1} \Rightarrow \{-1,+1\}$, $\limsup = 1$, $\liminf = -1$, diverges.

2.5 Cauchy sequences

DEF: Cauchy

$\forall\varepsilon > 0\ \exists N\ \forall m,n \ge N :\ |a_n - a_m| < \varepsilon$. In $\mathbb{R}$ (complete): **Cauchy $\iff$ convergent**. Use it to prove convergence *without knowing the limit*. **!** "$|a_{n+1} - a_n| \to 0$" is **not** enough ($a_n = \sqrt{n}$, $a_n = \sum 1/k$). You need control of $|a_n - a_m|$ for *all* $m, n \ge N$.

▶ RECIPE: show (a_n) is Cauchy

Typical: show $|a_{n+1} - a_n| \le Cq^n$ with $q < 1$. Then for $m > n$ $|a_m - a_n| \le \sum_{k=n}^{m-1} Cq^k \le \frac{Cq^n}{1-q} \to 0 \Rightarrow$ Cauchy $\Rightarrow$ convergent.

3 Series $\sum a_n$ and power series

3.1 Setup

DEF: series, absolute convergence

$\sum_{n=1}^\infty a_n := \lim_{N\to\infty} s_N$, $s_N = \sum_{n=1}^N a_n$ (a series IS the sequence of partial sums). **absolutely convergent:** $\sum |a_n| < \infty$. **abs. conv. $\Rightarrow$ conv.** (not conversely: $\sum \frac{(-1)^n}{n}$). **conditionally convergent:** converges but not absolutely.

3.2 Decision tree: does $\sum a_n$ converge?

▶ RECIPE: run these in order, stop at the first hit

1 $a_n \not\to 0$? $\Rightarrow$ **diverges**, done *(Trivialkriterium)*. (Always check this first, it is free.)
2 **Known shape?** geometric $\sum q^n$ conv. $\iff$ $|q| < 1$ (value $\frac{1}{1-q}$ from $n = 0$); p-series $\sum \frac{1}{n^p}$ conv. $\iff$ $p > 1$; $\sum \frac{1}{n \ln^p n}$ conv. $\iff$ $p > 1$.
3 **Factorials or c^n?** $\Rightarrow$ **ratio test** $\lim \left|\frac{a_{n+1}}{a_n}\right| =: q$. $q < 1$ conv. (abs.), $q > 1$ div., $q = 1$ useless $\Rightarrow$ next test.
4 **n-th powers ($a_n = b_n^n$, or n in the exponent)?** $\Rightarrow$ **root test** $\lim \sqrt[n]{|a_n|} =: q$, same rule. Root test is *stronger* than ratio.
5 **Looks like a known series?** $\Rightarrow$ **comparison** *(Majorante/Minorante)*: $0 \le a_n \le b_n$, $\sum b_n$ conv. $\Rightarrow \sum a_n$ conv.; $a_n \ge b_n \ge 0$, $\sum b_n$ div. $\Rightarrow \sum a_n$ div. Compare only *leading powers*: $\frac{3n^2+n}{n^4-1} \sim \frac{3}{n^2} \Rightarrow$ conv.
6 **Alternating $(-1)^n b_n$?** $\Rightarrow$ **Leibniz**: $b_n \ge 0$, *monotonically* decreasing, $b_n \to 0 \Rightarrow$ converges. (Remainder $|s - s_N| \le b_{N+1}$.)
7 **$a_n = f(n)$, $f \ge 0$ decreasing?** $\Rightarrow$ **integral test**: $\sum f(n)$ and $\int_1^\infty f\,dx$ share the same behaviour.
8 **Telescoping** $a_n = b_n - b_{n+1} \Rightarrow s_N = b_1 - b_{N+1}$; split by partial fractions first.

TRAP: test traps

- Ratio/root $q = 1$ gives **no information** ($\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ both have $q = 1$).
- Leibniz needs *monotone* b_n — $b_n = \frac{2+(-1)^n}{n} \to 0$ but not monotone.
- Comparison needs $a_n \ge 0$; otherwise apply it to $|a_n|$ ($\Rightarrow$ absolute convergence).
- Convergence is unaffected by finitely many terms / by the starting index.
- $\sum a_n$ conv. and $\sum b_n$ conv. $\Rightarrow \sum(a_n + b_n)$ conv. But conv. $\times$ conv. termwise says nothing.

EXAMPLE: run the tree on five series

- $\sum \frac{n^2}{2^n}$: $c^n \Rightarrow$ ratio $= \frac{(n+1)^2}{2n^2} \to \frac{1}{2} < 1 \Rightarrow$ **conv.**
- $\sum \frac{n!}{n^n}$: ratio $= \frac{(n+1)!n^n}{n!(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1 \Rightarrow$ **conv.**
- $\sum \left(\frac{n}{2n+1}\right)^n$: n-th power $\Rightarrow$ root $= \frac{n}{2n+1} \to \frac{1}{2} < 1 \Rightarrow$ **conv.**
- $\sum \frac{2n+3}{n^2+7}$: leading behaviour $\frac{2}{n} \Rightarrow$ compare with $\sum \frac{1}{n} \Rightarrow$ **div.**
- $\sum \frac{(-1)^n}{\sqrt{n}}$: $\frac{1}{\sqrt{n}} \downarrow 0 \Rightarrow$ Leibniz $\Rightarrow$ conv.; but $\sum \frac{1}{\sqrt{n}}$ div. $\Rightarrow$ **conditionally conv.**

FACT: rearrangement and products

- **Riemann rearrangement:** conditionally convergent $\Rightarrow$ reordering can produce *any* value in $\mathbb{R}\cup\{\pm\infty\}$.
- **Dirichlet:** absolutely convergent $\Rightarrow$ every reordering gives the same sum.
- **Cauchy product** $c_n = \sum_{k=0}^n a_k b_{n-k}$: if at least one of the two converges absolutely then $\sum c_n = (\sum a_n)(\sum b_n)$ (Mertens).

FACT: values to know

$\sum_{n\ge0} q^n = \frac{1}{1-q}$, $\sum_{n\ge1} nq^{n-1} = \frac{1}{(1-q)^2}$ ($|q| < 1$); $\sum_{n\ge0} \frac{x^n}{n!} = e^x$; $\sum_{n\ge1} \frac{(-1)^{n+1}}{n} = \ln 2$; $\sum_{n\ge1} \frac{1}{n^2} = \frac{\pi^2}{6}$; $\sum_{n\ge1} \frac{1}{n(n+1)} = 1$.

3.3 Power series $\sum a_n(x - x_0)^n$

▶ RECIPE: "for which x does it converge?"

1 Read off the centre x_0 (in $(x-2)^n$: $x_0 = 2$) and the coefficient a_n (everything without $(x-x_0)^n$).
2 **Radius:** $\frac{1}{R} = \lim \left|\frac{a_{n+1}}{a_n}\right|$ (factorials!) or $\frac{1}{R} = \limsup \sqrt[n]{|a_n|}$ (powers). $R = \infty$ if the limit is 0; $R = 0$ if it is ∞.
3 Converges absolutely on $|x - x_0| < R$, diverges on $|x - x_0| > R$.
4 **Endpoints $x = x_0 \pm R$ separately!** Plug them in and use §3.2 (usually Leibniz or p-series).
5 Answer as an interval, e.g. $[x_0 - R,\ x_0 + R)$.

EXAMPLE: $\sum_{n\ge1} \frac{n^n(x-2)^n}{(2n)!}$

$a_n = \frac{n^n}{(2n)!}$ (factorial $\Rightarrow$ ratio test). $\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^{n+1}}{(2n+2)!}\cdot\frac{(2n)!}{n^n} = \frac{(n+1)(1+\frac{1}{n})^n}{(2n+1)(2n+2)} \xrightarrow[n\to\infty]{} \frac{e\,(n+1)}{4n^2} \to 0$. So $1/R = 0$, $R = \infty$: **converges for all $x \in \mathbb{R}$.**

FACT: what power series give you for free

Inside $|x - x_0| < R$: the sum is C^∞, and you may **differentiate and integrate termwise**; R does not change. Convergence is *uniform* on every closed $[x_0 - r, x_0 + r]$ with $r < R$ (not necessarily on the whole open interval). Coefficients are unique: $a_n = \frac{f^{(n)}(x_0)}{n!}$.

4 Functions: limits, continuity, function sequences (f_n)

4.1 Basic properties of a function

▶ RECIPE: injective / surjective / bijective / inverse

1 **Injective:** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Shortcut: *strictly* monotone $\Rightarrow$ injective (show $f' > 0$ or $f' < 0$).
2 **Surjective onto** Y: solve $f(x) = y$ for arbitrary $y \in Y$; or show $\operatorname{im} f = Y$ via IVT + the limits at the ends of the domain.
3 **Bijective** = both $\Rightarrow f^{-1}$ exists. Compute it by solving $y = f(x)$ for x and swapping names.
4 **f^{-1} is continuous** if f is continuous and strictly monotone on an interval; and differentiable with $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$ provided $f'(x_0) \ne 0$.

Also: **even** $f(-x) = f(x)$, **odd** $f(-x) = -f(x)$; **periodic** $f(x+T) = f(x)$; **bounded** $|f| \le C$.

▶ RECIPE: "determine a, b so that f is continuous / differentiable"

1 Continuity at the seam x_0: set $\lim_{x\to x_0^-} f = \lim_{x\to x_0^+} f = f(x_0) \Rightarrow$ one equation.
2 Differentiability: additionally $\lim_{x\to x_0^-} f' = \lim_{x\to x_0^+} f' \Rightarrow$ second equation.
3 Solve the (linear) system for a, b. Two unknowns need exactly these two conditions.

4.2 Limits of functions

DEF: ε–δ and one-sided

$\lim_{x\to x_0} f(x) = L :\iff \forall\varepsilon > 0\ \exists\delta > 0 :\ 0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon$. Limit exists $\iff$ $\lim_{x\to x_0^-} f = \lim_{x\to x_0^+} f$ (and both finite). $\lim_{x\to\infty} f = L$: $\forall\varepsilon\ \exists K :\ x > K \Rightarrow |f(x) - L| < \varepsilon$.

▶ RECIPE: compute $\lim_{x\to x_0} f(x)$

1 Plug in x_0. No problem $\Rightarrow$ done (continuity).
2 $\frac{0}{0}$: factor & cancel, expand roots by the conjugate, or $\Rightarrow$ Taylor (§5.6) / l'Hospital (§5.5).
3 $\frac{c\ne0}{0}$: $\to \pm\infty$; determine the sign from the left/right $\Rightarrow$ answer only exists one-sided.
4 $x \to \pm\infty$ rational: divide by highest power in the denominator. $\sqrt{\cdot}$ for $x \to -\infty$: $\sqrt{x^2} = |x| = -x$. **!**
5 $|x|$, sgn, piecewise, floor: **always** check both one-sided limits.
6 Bounded $\times$ null: squeeze, e.g. $x \sin\frac{1}{x} \to 0$ but $\sin\frac{1}{x}$ has no limit.

4.3 Continuity

DEF: continuity + sequential criterion

f continuous at $x_0 :\iff \forall\varepsilon > 0\ \exists\delta > 0 :\ |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$ $\iff$ $\lim_{x\to x_0} f(x) = f(x_0)$ $\iff$ **(Heine)** for every sequence $x_n \to x_0$ one has $f(x_n) \to f(x_0)$.

▶ RECIPE: prove / disprove continuity at x_0

Prove: built from $+, -, \times, \div$ (den. $\ne 0$), $\circ$ of continuous standard functions $\Rightarrow$ continuous, one line. Otherwise ε–δ: bound $|f(x) - f(x_0)| \le C|x - x_0|$ and take $\delta = \varepsilon/C$.
Disprove: find *two* sequences $x_n \to x_0$, $y_n \to x_0$ with $\lim f(x_n) \ne \lim f(y_n)$ (Heine). Classic: $x_n = \frac{1}{2\pi n}$ vs. $y_n = \frac{1}{\pi/2+2\pi n}$ for $\sin\frac{1}{x}$; rationals vs. irrationals for Dirichlet.

FACT: types of discontinuity (Unstetigkeitsstellen)

removable *(hebbar)*: both one-sided limits exist and are equal $\ne f(x_0)$. **jump** *(Sprung)*: both exist, differ (this is what monotone functions have; at most countably many). **essential/oscillating**: at least one one-sided limit fails to exist ($\sin\frac{1}{x}$, $\frac{1}{x}$).

4.4 The three big theorems on $[a, b]$ — and when to use which

FACT: cite these verbatim

$f : [a, b] \to \mathbb{R}$ **continuous**, $[a, b]$ compact:

- **IVT** *(Zwischenwertsatz)*: c between $f(a)$ and $f(b) \Rightarrow \exists x \in [a, b] : f(x) = c$. *Use for: existence of a zero / solution / fixed point.*
- **Extreme value thm** *(Satz vom Max/Min, Weierstrass)*: f is bounded and *attains* max and min. *Use for: "the maximum is attained", bounding.*
- **Heine:** f continuous on a **compact** interval $\Rightarrow f$ is *uniformly* continuous. *Use to get uniform continuity for free.*

! All three fail on open/unbounded intervals ($f(x) = 1/x$ on $(0, 1]$).

▶ RECIPE: show "$f(x) = g(x)$ has a solution" / "f has a zero"

1 Set $h := f - g$ (fixed point $f(x) = x$: $h(x) := f(x) - x$). State: h continuous as difference of continuous functions.
2 Find two points with opposite signs: evaluate at convenient points, or take limits $x \to \pm\infty$ and argue with the sign for large $|x|$.
3 IVT $\Rightarrow \exists x^*$ with $h(x^*) = 0$. **Existence done.**
4 **Uniqueness:** show h is strictly monotone ($h' > 0$ or $h' < 0$ everywhere) $\Rightarrow$ at most one zero. Alternative: Rolle — two zeros would force $h'(\xi) = 0$, contradiction.

EXAMPLE: $e^x = p(x)$ — typical multiple choice

- IVT gives a solution **not always**: $p(x) = -1$ has none ($e^x > 0$).
- p of *odd* degree $\Rightarrow$ always at least one solution ($e^x - p(x)$ changes sign).
- Number of solutions is always *finite*? **Yes** — $e^x - p(x)$ is analytic, $\neq 0$, so finitely many zeros; and for any N there is a polynomial with $\geq N$ solutions (take the N-th Taylor polynomial of e^x-like constructions / oscillating differences).

4.5 Uniform continuity (gleichmässige Stetigkeit)

DEF: difference to ordinary continuity

continuous: $\forall x_0 \forall \varepsilon \exists \delta(\varepsilon, x_0)$. **uniformly continuous:** $\forall \varepsilon\ \exists \delta(\varepsilon)\ \forall x, y : |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.
The δ must not depend on the point. unif. cont. $\Rightarrow$ cont., never the other way.

RECIPE: "is f uniformly continuous?"

1 **Domain compact** $[a,b]$ **and** f **continuous?** $\Rightarrow$ YES by Heine. Done, one line.
2 **Is** f' **bounded on the domain,** $|f'| \leq L$? $\Rightarrow$ MVT gives $|f(x) - f(y)| \leq L|x-y|$ (Lipschitz) $\Rightarrow$ YES, take $\delta = \varepsilon / L$. *This is the workhorse on* $\mathbb{R}$.
3 f **continuous on** $\mathbb{R}$ **with finite limits at** $\pm\infty$? $\Rightarrow$ YES (compactness argument on a large interval + almost constant outside).
4 **Suspect NO?** Find x_n, y_n with $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0 > 0$. Classics: x^2 on $\mathbb{R}$ ($x_n = n$, $y_n = n + \frac{1}{n}$); $\frac{1}{x}$ on $(0,1)$; $\sin\frac{1}{x}$ on $(0,1)$.
5 **Case distinction** if a parameter can degenerate (e.g. $n = 0$ vs. $n \geq 1$) — treat the degenerate case separately.

! Unbounded f' does NOT prove "not uniformly continuous": $\sqrt{x}$ on $[0,1]$ has $f' \to \infty$ but is unif. cont. (compact).

EXAMPLE: $f_n(x) = \dfrac{x}{1+nx^2}$ on $\mathbb{R}$: uniformly continuous?

$n = 0$: $f_0(x) = x$, Lipschitz with $L = 1 \Rightarrow$ unif. cont.

$n \geq 1$: $f_n'(x) = \dfrac{1 - nx^2}{(1+nx^2)^2}$, so $|f_n'(x)| \leq \dfrac{1+nx^2}{(1+nx^2)^2} = \dfrac{1}{1+nx^2} \leq 1$.

Bounded derivative $\Rightarrow$ Lipschitz with $L = 1 \Rightarrow$ **uniformly continuous for every** n.

EXAMPLE: disproving uniform continuity: $f(x) = x^2$ on $\mathbb{R}$

Take $x_n = n$, $y_n = n + \frac{1}{n}$. Then $|x_n - y_n| = \frac{1}{n} \to 0$, but $|f(x_n) - f(y_n)| = \left|n^2 - \left(n + \frac{1}{n}\right)^2\right| = 2 + \frac{1}{n^2} \geq 2 =: \varepsilon_0$. So no δ can work for $\varepsilon = 2 \Rightarrow$ **not uniformly continuous**. (Same pattern for $\frac{1}{x}$ on $(0,1)$: $x_n = \frac{1}{n}$, $y_n = \frac{1}{2n}$ $\Rightarrow$ difference n.)

4.6 Sequences of functions (f_n) — pointwise limit

RECIPE: find the pointwise limit $f(x) = \lim_n f_n(x)$

1 **Freeze** x (treat it as a constant), let $n \to \infty$ using §2.2.
2 Do a **case distinction in** x whenever the answer changes: typically $x = 0$ vs. $x \neq 0$, $|x| < 1$ vs. $= 1$ vs. > 1.
3 Write the limit function piecewise. **If f is discontinuous, convergence cannot be uniform** (see §4.7).

EXAMPLE: pointwise limits you should recognise

$\dfrac{x}{1+nx^2}$: $x = 0 \Rightarrow 0$; $x \neq 0 \Rightarrow \frac{x}{nx^2} \to 0$. So $f \equiv 0$ on $\mathbb{R}$.
x^n on $[0,1] \Rightarrow f = 0$ on $[0,1)$, $f(1) = 1$ (discontinuous $\Rightarrow$ not uniform).
$\frac{\sin(nx)}{n} \to 0$; $nxe^{-nx^2} \to 0$; $\arctan(nx) \to \frac{\pi}{2}\operatorname{sgn}(x)$.

4.7 Uniform convergence $f_n \rightrightarrows f$

DEF: the only formula you need

$f_n \to f$ **uniformly on** $D :\iff \|f_n - f\|_\infty := \sup_{x \in D} |f_n(x) - f(x)| \xrightarrow[n\to\infty]{} 0$ $\iff \forall \varepsilon \exists N \forall n \geq N\ \forall x \in D : |f_n(x) - f(x)| < \varepsilon$ (one N for *all* x).

RECIPE: decide uniform convergence (the standard 3-point task)

1 Get the pointwise limit f first (§4.6). Uniform convergence can only be towards it.
2 Form $g_n(x) := |f_n(x) - f(x)|$ and compute $M_n := \sup_x g_n(x)$: differentiate g_n w.r.t. x (n fixed), set $g_n'(x) = 0$, plug the critical point back in. Also check the boundary/$\pm\infty$.
3 $M_n \to 0 \Rightarrow$ **uniform**. $M_n \not\to 0 \Rightarrow$ **only pointwise**.
4 **Shortcut for NO:** exhibit one sequence x_n with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0 > 0$ (usually x_n = the maximiser). One such sequence is a complete disproof.
5 **Shortcut for NO:** f_n all continuous but f is not $\Rightarrow$ not uniform (contrapositive of the Uniform Limit Theorem).

EXAMPLE: $f_n(x) = \frac{x}{1+nx^2}$ on $\mathbb{R}$: uniform?

$f \equiv 0$, so $M_n = \sup_x \left|\frac{x}{1+nx^2}\right|$. $f_n'(x) = \frac{1-nx^2}{(1+nx^2)^2} = 0 \Rightarrow x = \pm\frac{1}{\sqrt{n}}$. $f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{2} = \frac{1}{2\sqrt{n}}$. So $M_n = \frac{1}{2\sqrt{n}} \to 0 \Rightarrow$ **converges uniformly to 0 on all of** $\mathbb{R}$.

EXAMPLE: $f_n(x) = nxe^{-nx^2}$ on $[0,1]$: not uniform

Pointwise $f \equiv 0$. Maximiser: $x_n = \frac{1}{\sqrt{2n}}$, $f_n(x_n) = \sqrt{\frac{n}{2}}e^{-1/2} \to \infty \neq 0 \Rightarrow$ **only pointwise**.

FACT: what uniform convergence buys you (and what it does not)

f_n continuous on D, $f_n \rightrightarrows f \Rightarrow$

- f **is continuous** (Uniform Limit Theorem). *Pointwise alone gives nothing* — x^n *on* $[0,1]$.
- On $[a,b]$: f is Riemann-integrable and $\int_a^b f_n \to \int_a^b f$ (swap lim and $\int$). *Pointwise alone: false.*
- **! Derivatives do NOT pass to the limit:** $\frac{\sin(nx)}{n} \rightrightarrows 0$ but $f_n' = \cos(nx)$ diverges. You need $f_n' \rightrightarrows g$ *uniformly* plus convergence at one point; then $f' = g$.
- $\sup|f_n| \geq \frac{1}{2n}$ does *not* block uniform convergence to 0 — $\frac{1}{2n} \to 0$. Only a lower bound that does *not* vanish blocks it.
- max attained at a different point x_n each time is irrelevant; only the *value* M_n counts.

FACT: Weierstrass M-test (for $\sum f_n$)

$|f_n(x)| \leq M_n\ \forall x \in D$ and $\sum M_n < \infty \Rightarrow \sum f_n$ converges **absolutely and uniformly** on D. Use for function series / power series on $[-r, r]$.

5 Derivatives and their applications

5.1 Differentiability

DEF: three equivalent formulations

$f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$ (**two-sided**, finite) $\iff \exists K \in \mathbb{R} : f(x) = f(x_0) + K(x - x_0) + o(|x - x_0|)$ as $x \to x_0$, and then $K = f'(x_0)$.
diff. $\Rightarrow$ **continuous**; continuous $\not\Rightarrow$ diff. ($|x|$ at 0).

TRAP: Landau o / O — the classic multiple-choice trap

$f = O(g)$: $|f| \leq C|g|$ near x_0 (*same order or smaller*). $f = o(g)$: $f/g \to 0$ (*strictly smaller*).

- $f(x) - f(x_0) - K(x - x_0) = o(|x - x_0|) \Rightarrow$ **differentiable**. Note $(x - x_0)$, *not* Kx: with Kx the statement is false/meaningless.
- $f(x) - f(x_0) = O(|x - x_0|) \Rightarrow$ only Lipschitz near $x_0 \Rightarrow$ **not** differentiable ($|x|$ satisfies it).
- Only $h \to 0^+$ exists $\Rightarrow$ only *right*-differentiable $\Rightarrow$ **not** differentiable.
- "for an arbitrary polynomial p" $\Rightarrow$ useless, p must be the linear Taylor part.

RECIPE: piecewise function: differentiable at the seam x_0?

1 Continuity first: $\lim_{x\to x_0^-} f = \lim_{x \to x_0^+} f = f(x_0)$. Fails $\Rightarrow$ not differentiable, done.
2 Compute the one-sided *difference quotients* (not just $\lim f'$) and compare. Equal $\Rightarrow$ differentiable with that value.
3 For C^1: additionally $\lim_{x\to x_0^-} f'(x) = \lim_{x \to x_0^+} f'(x)$.

Ex. $f(x) = x^2 \sin\frac{1}{x}$ ($x \neq 0$), $f(0) = 0$: $f'(0) = \lim h \sin\frac{1}{h} = 0$ exists, but f' is not continuous at $0 \Rightarrow$ diff. but not C^1.

5.2 Rules — and how not to misuse them

FACT: the five rules

$(f \pm g)' = f' \pm g'$; $(fg)' = f'g + fg'$; $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$; **chain** $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$; **inverse** $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$ with $y_0 = f(x_0)$.
Logarithmic differentiation for $u(x)^{v(x)}$: write $u^v = e^{v \ln u} \Rightarrow (u^v)' = u^v\left(v' \ln u + \frac{vu'}{u}\right)$.

TRAP: using the derivative/antiderivative table (§9) correctly

The table lists $F(x)$ for the *bare* argument x. If the argument is $ax + b$:

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C \qquad \text{(divide by the inner derivative)}$$

! This works **only for linear inner functions**. For anything else substitute properly.
Worked trap: $\int \frac{1}{2\cos^2(\frac{1}{2}x)}\,dx$. Table: $\int \frac{1}{\cos^2 u}\,du = \tan u$. Inner $u = \frac{1}{2}x$ $\Rightarrow u' = \frac{1}{2} \Rightarrow$ divide by $\frac{1}{2}$, i.e. *multiply* by 2, and keep the prefactor $\frac{1}{2}$: $\frac{1}{2} \cdot 2\tan(\frac{1}{2}x) = \tan\left(\frac{x}{2}\right) + C$. **Always verify by differentiating your answer** — 5 seconds, catches every chain-rule slip.

5.3 Mean value theorems (Mittelwertsätze)

FACT: Fermat / Rolle / Lagrange / Cauchy

- **Fermat:** x_0 *interior* local extremum and f diff. there $\Rightarrow f'(x_0) = 0$. (Converse false: x^3.)
- **Rolle:** f cont. on $[a,b]$, diff. on (a,b), $f(a) = f(b) \Rightarrow \exists \xi \in (a,b) : f'(\xi) = 0$.
- **MVT (Lagrange):** same hypotheses $\Rightarrow \exists \xi \in (a,b) : f'(\xi) = \frac{f(b) - f(a)}{b-a}$.
- **Cauchy MVT:** $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$ (this is what proves l'Hospital).

RECIPE: typical MVT uses

- **Prove an inequality** $|f(x) - f(y)| \leq L|x - y|$: MVT + bound $|f'| \leq L$.
- **Show f is constant:** $f' \equiv 0$ on an interval $\Rightarrow$ MVT $\Rightarrow$ constant.
- **Count zeros:** n zeros of $f \Rightarrow \geq n - 1$ zeros of f' (Rolle between consecutive zeros). Contrapositive: f' has $\leq k$ zeros $\Rightarrow f$ has $\leq k + 1$ zeros.
- **Compare two functions:** $g := f - h$, show $g(a) = 0$ and $g' > 0 \Rightarrow f > h$ on (a, ∞).

5.4 Curve discussion (Kurvendiskussion) / extrema

RECIPE: find and classify all extrema of f on I

1 **Domain** + points where f is not differentiable (kinks, poles, boundary of I) — these are extremum candidates too.
2 **Critical points:** solve $f'(x) = 0$.
3 **Classify:** $f''(x_0) > 0 \Rightarrow$ local **min**; $f''(x_0) < 0 \Rightarrow$ local **max**; $f''(x_0) = 0 \Rightarrow$ inconclusive, use the *sign change of* f' ($- \to +$ min, $+ \to -$ max, no change $\Rightarrow$ saddle/inflection).
4 **Global on** $[a,b]$: compare f at all critical points *and* at a, b. Largest value = global max (exists by Weierstrass).
5 **Global on open/unbounded:** additionally take $\lim_{x \to \partial I} f$; if the limit exceeds all candidate values, the extremum is not attained.
6 **Monotonicity:** sign table of f'. **Convexity:** sign table of f''; sign change of f'' = **inflection point** *(Wendepunkt)*.
7 Asymptotes: $\lim_{x\to\pm\infty}(f(x) - (mx + c)) = 0$ with $m = \lim \frac{f(x)}{x}$, $c = \lim(f(x) - mx)$.

FACT: convexity (konvex) — exam favourite

f convex on $I \iff f'' \geq 0 \iff f'$ increasing $\iff f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) \iff$ graph lies above every tangent.

- Convex + $f'(x_0) = 0$ at an interior point $\Rightarrow x_0$ is a **global minimum**. *(True statement in MC questions.)*
- A convex differentiable function cannot have several isolated critical points (the critical set is an interval).
- Its global *maximum* on $[a,b]$ is always attained at a **boundary** point.
- Its global *minimum* may well be interior — so "min only on the boundary" is false.

EXAMPLE: full curve discussion of $f(x)=\dfrac{x}{1+x^2}$

Domain $\mathbb{R}$, odd. $f'=\dfrac{1-x^2}{(1+x^2)^2}\Rightarrow$ zeros $x=\pm1$.

$f''=\dfrac{2x(x^2-3)}{(1+x^2)^3}\Rightarrow f''(1)=-\frac12<0\Rightarrow$ **max** $f(1)=\frac12$; $f''(-1)>0\Rightarrow$ **min** $f(-1)=-\frac12$.

Inflection $x=0,\pm\sqrt3$ (sign of f'' changes). **Monotone** incr. on $[-1,1]$, decr. outside. **Asymptote** $y=0$ since $f\to0$ for $x\to\pm\infty$. Global max/min $=\pm\frac12$ (attained).

5.5 l'Hospital (Bernoulli–de l'Hospital)

▸ RECIPE: use it safely

1. **Check the form first:** only $\frac00$ or $\frac{\pm\infty}{\pm\infty}$. Writing it down without checking costs points.
2. Differentiate *numerator and denominator separately* (**!** not the quotient rule).
3. Repeat if the form persists. If it never resolves or loops ($\frac{e^x}{e^x}$-style, $\frac{x}{\sqrt{1+x^2}}$), switch method.
4. Verify the new limit exists; if it does not, l'Hospital says **nothing** (e.g. $\frac{x+\sin x}{x}$).

Reduce other forms first: $0\cdot\infty$: $fg=\frac{f}{1/g}$. $\infty-\infty$: common denominator or conjugate. $1^\infty,0^0,\infty^0$: $u^v=e^{v\ln u}$, do l'Hospital in the exponent, exponentiate at the end.

EXAMPLE: the indeterminate forms, one line each

- $\lim_{x\to0}\frac{1-\cos x}{x^2}\overset{\text{Taylor}}{=}\lim\frac{x^2/2}{x^2}=\frac12$.
- $\lim_{x\to0^+}x\ln x=\lim\frac{\ln x}{1/x}\overset{\infty/\infty}{=}\lim\frac{1/x}{-1/x^2}=\lim(-x)=0$.
- $\lim_{x\to0^+}x^x=e^{\lim x\ln x}=e^0=1$.
- $\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^x=e^{\lim x\ln(1+a/x)}=e^a$.
- $\lim_{x\to0}\left(\frac{1}{\sin x}-\frac1x\right)=\lim\frac{x-\sin x}{x\sin x}=\lim\frac{x^3/6}{x^2}=0$.
- $\lim_{x\to\infty}\frac{x+\sin x}{x}=1$ — **!** l'Hospital gives $1+\cos x$ (no limit) and is *inapplicable*; just divide by x.

5.6 Taylor — the best limit tool

FACT: Taylor formula

$f(x)=\sum_{k=0}^n\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k+R_n(x)$, **Lagrange remainder** $R_n(x)=\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$ for some ξ between x_0 and x; also $R_n=o(|x-x_0|^n)$.

Error bound: $|R_n(x)|\le\frac{\max_\xi|f^{(n+1)}(\xi)|}{(n+1)!}|x-x_0|^{n+1}$ — that is the whole "estimate the error" task.

▸ RECIPE: limits $x\to0$ via Taylor (faster and safer than l'Hospital)

1. Look at the denominator's power x^m. You need every series up to order m (one more if things cancel).
2. Insert the standard series (§9), expand, cancel.
3. The surviving lowest-order term gives the limit; everything else is $o(x^m)$.

EXAMPLE: $\lim\limits_{x\to0}\dfrac{\sin x-x+\frac{x^3}{6}}{x^5}$

$\sin x=x-\frac{x^3}{6}+\frac{x^5}{120}-\frac{x^7}{5040}+\dots$

Numerator $=\frac{x^5}{120}+O(x^7)\Rightarrow$ limit $=\boxed{\frac{1}{120}}$.

(l'Hospital would need **five** differentiations — don't.)

▸ RECIPE: "approximate $f(x)$ and bound the error"

1. Choose x_0 near x where you know f exactly $(0,1,\frac{\pi}{6},\dots)$ and the order n.
2. Write $T_n(x)=\sum_{k\le n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ and evaluate.
3. Error: $|R_n|\le\frac{M}{(n+1)!}|x-x_0|^{n+1}$ with $M:=\max|f^{(n+1)}|$ on the interval between x_0 and x.
4. If a target accuracy is given, solve that bound $<\varepsilon$ for n.

Ex. $e^{0.1}$, $n=3$: $T_3=1+0.1+0.005+\frac{0.001}{6}\approx1.10517$; $M=e^{0.1}<1.2\Rightarrow|R_3|\le\frac{1.2}{24}10^{-4}=5\cdot10^{-6}$.

FACT: Newton's method (if it appears)

$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$; converges quadratically near a simple root if $f'\ne0$ and $f\in C^2$. Geometrically: the zero of the tangent at x_n.

FACT: Taylor series to know by heart (all around $x_0=0$)

$e^x=\sum\frac{x^n}{n!}$ $(R=\infty)$; $\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}\mp$; $\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}\mp$;
$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}\mp$ $(|x|<1)$; $\frac{1}{1-x}=\sum x^n$; $(1+x)^\alpha=\sum\binom{\alpha}{n}x^n$;
$\arctan x=x-\frac{x^3}{3}+\frac{x^5}{5}\mp$; $\sinh x=x+\frac{x^3}{3!}+\dots$; $\cosh x=1+\frac{x^2}{2!}+\dots$

Tricks: substitute (e^{-x^2}), differentiate/integrate termwise, multiply series — do *not* recompute derivatives.

6 Integration

6.1 Decision tree: how do I integrate this?

▸ RECIPE: look at the integrand, take the first match

1. **In the table (§9)?** $\Rightarrow$ read it off. Inner function $ax+b\Rightarrow$ multiply by $\frac1a$. **!** nothing else may be "inner".
2. $\int f'(x)\,g(f(x))\,dx$**, i.e. the inner derivative is a factor?** $\Rightarrow$ **substitution** $u=f(x)$. Special case $\int\frac{f'}{f}\,dx=\ln|f|+C$ and $\int f^nf'\,dx=\frac{f^{n+1}}{n+1}$.
3. **Product of two unrelated types** (poly $\times$ e^x/sin/ln/arctan) $\Rightarrow$ **by parts**.
4. **Rational function** $\frac pq\Rightarrow$ polynomial division if $\deg p\ge\deg q$, then **partial fractions**.
5. **Rational in** $\sin,\cos\Rightarrow$ **Weierstrass** $t=\tan\frac x2$; first try half-angle identities, often much shorter.
6. $\sqrt{a^2-x^2}$, $\sqrt{a^2+x^2}$, $\sqrt{x^2-a^2}\Rightarrow$ **trig substitution** $x=a\sin\theta$ / $a\tan\theta$ / $a\sec\theta$.
7. $\sqrt[n]{ax+b}\Rightarrow$ substitute $u=\sqrt[n]{ax+b}$ (kills the root completely).
8. $\sin^m x\cos^n x\Rightarrow$ odd power: peel one factor off, substitute the other. Both even: half-angle formulas.
9. **Nothing works** $\Rightarrow$ symmetry (§6.9), or the answer contains a non-elementary integral like $\int\frac{\sin x}{x}\,dx$ — that is allowed in MC answers.

ALWAYS finish by differentiating your result. That is the only real check under time pressure.

6.2 Substitution (Substitution)

▸ RECIPE: step by step (both directions)

1. Choose $u=g(x)$ (usually the inner function of a composition).
2. Compute $du=g'(x)\,dx\Rightarrow$ solve for $dx=\frac{du}{g'(x)}$ and make *every* x disappear.
3. **Definite integral:** convert the bounds $a\mapsto g(a)$, $b\mapsto g(b)$ — then you must *not* substitute back.
4. **Indefinite integral:** integrate in u, then **substitute** $u=g(x)$ **back**.
5. **!** After back-substitution, differentiate: the chain rule factor $g'(x)$ must cancel. This is where most marks are lost.

Reverse direction ($x=\varphi(t)$, needs φ bijective/C^1): $\int f(x)\,dx=\int f(\varphi(t))\varphi'(t)\,dt$, back via $t=\varphi^{-1}(x)$.

TRAP: the chain-rule-back-substitution trap

$\int\frac{1}{2\cos^2(\frac x2)}\,dx$: table gives $\int\frac{du}{\cos^2u}=\tan u$. Substitute $u=\frac x2$, $dx=2\,du$: $\int\frac{2\,du}{2\cos^2u}=\tan u=\tan\frac x2+C$. **NOT** $2\tan\frac x2$. The factor 2 from dx and the $\frac12$ in front cancel — write the substitution out, never patch factors by feeling.

6.3 Integration by parts (partielle Integration)

▸ RECIPE: $\int u\,v'\,dx=uv-\int u'v\,dx$

1. Choose $u=$ the part that gets *simpler* when differentiated. Priority **LIATE**: **L**og, **I**nverse trig, **A**lgebraic (poly), **T**rig, **E**xponential — earlier in the list becomes u.
2. $v'=$ the rest; integrate it to get v (no $+C$ needed here).
3. Apply the formula. Repeat if the new integral is still a product (x^2e^x: twice).
4. $\int\ln x\,dx$, $\int\arctan x\,dx$: take $v'=1$. $\Rightarrow x\ln x-x$, $x\arctan x-\frac12\ln(1+x^2)$.
5. **Cycle trick** ($\int e^x\sin x\,dx$): after two rounds the original integral I reappears $\Rightarrow$ solve $I=\dots-I$ algebraically for I, divide by 2.

EXAMPLE: multiple choice "$\int x\ln x\sin x\,dx=\dots$"

Do **not** integrate — **differentiate the four options** and compare with the integrand. 20 s instead of 5 min.
Check of $-x\ln x\cos x+\ln x\sin x+\sin x-\int\frac{\sin x}{x}\,dx$:
$\frac{d}{dx}$: $-(\ln x+1)\cos x+x\ln x\sin x+\frac{\sin x}{x}+\ln x\cos x+\cos x-\frac{\sin x}{x}=x\ln x\sin x$ ✓

6.4 Partial fractions (Partialbruchzerlegung)

▸ RECIPE: $\int\frac{p(x)}{q(x)}\,dx$

1. $\deg p\ge\deg q\Rightarrow$ polynomial division first.
2. Factor q completely over $\mathbb{R}$ (linear factors + irreducible quadratics).
3. Ansatz: $(x-a)\Rightarrow\frac{A}{x-a}$; $(x-a)^k\Rightarrow\frac{A_1}{x-a}+\dots+\frac{A_k}{(x-a)^k}$; irreducible $x^2+bx+c\Rightarrow\frac{Ax+B}{x^2+bx+c}$ (and powers of it analogously).
4. Multiply out, compare coefficients *or* plug in the roots (**cover-up**: fastest for simple linear factors).
5. Integrate: $\int\frac{A}{x-a}=A\ln|x-a|$; $\int\frac{A}{(x-a)^k}=\frac{A}{(1-k)(x-a)^{k-1}}$; $\int\frac{Ax+B}{x^2+bx+c}\Rightarrow$ split into $\frac A2\cdot\frac{2x+b}{x^2+bx+c}$ (gives ln) + rest $\Rightarrow$ complete the square $\Rightarrow\frac1a\arctan\frac{x-p}{a}$.

EXAMPLE: four integrals, fully worked

(a) **Substitution.** $\int\frac{x}{1+x^2}\,dx$: $u=1+x^2$, $du=2x\,dx\Rightarrow\frac12\int\frac{du}{u}=\frac12\ln(1+x^2)+C$.
(b) **By parts, twice.** $\int x^2e^x\,dx$: $u=x^2,v'=e^x\Rightarrow x^2e^x-2\int xe^x\,dx$; again $\Rightarrow x^2e^x-2(xe^x-e^x)=e^x(x^2-2x+2)+C$.
(c) **Partial fractions.** $\int\frac{1}{x^2-1}\,dx=\int\left(\frac{1/2}{x-1}-\frac{1/2}{x+1}\right)dx=\frac12\ln\left|\frac{x-1}{x+1}\right|+C$.
(d) **Trig substitution.** $\int\sqrt{1-x^2}\,dx$: $x=\sin\theta$, $dx=\cos\theta\,d\theta\Rightarrow\int\cos^2\theta\,d\theta=\frac\theta2+\frac{\sin2\theta}{4}\Rightarrow$ back: $\frac12\arcsin x+\frac12x\sqrt{1-x^2}+C$ (using $\sin2\theta=2x\sqrt{1-x^2}$). **!** back-substitution needs the identity, not just $\theta=\arcsin x$.

6.5 Rational in sin / cos

▸ RECIPE: half-angle first, Weierstrass only if needed

Try first: $1+\cos x=2\cos^2\frac x2$, $1-\cos x=2\sin^2\frac x2$, $\sin x=2\sin\frac x2\cos\frac x2$.
Weierstrass $t=\tan\frac x2$: $\sin x=\frac{2t}{1+t^2}$, $\cos x=\frac{1-t^2}{1+t^2}$, $dx=\frac{2}{1+t^2}dt\Rightarrow$ rational in $t\Rightarrow$ §6.4.

EXAMPLE: $\int\frac{1}{1+\cos x}\,dx$

$1+\cos x=2\cos^2\frac x2\Rightarrow\int\frac{dx}{2\cos^2\frac x2}\overset{u=x/2}{=}\int\frac{2\,du}{2\cos^2u}=\tan u=\boxed{\tan\frac{x}{2}+C}$.

Check: $\frac{d}{dx}\tan\frac x2=\frac{1}{2\cos^2\frac x2}=\frac{1}{1+\cos x}$ ✓
Twin: $\int\frac{dx}{1-\cos x}=-\cot\frac x2+C$.

6.6 Definite integrals, FTC, symmetry

FACT: Fundamental theorem (Hauptsatz)

f continuous on $[a,b]$, $F(x):=\int_a^xf(t)\,dt\Rightarrow F$ differentiable and $F'=f$.
Newton–Leibniz: $\int_a^bf=F(b)-F(a)$ for any antiderivative F.
Variable bounds (chain rule!): $\frac{d}{dx}\int_{u(x)}^{v(x)}f(t)\,dt=f(v(x))v'(x)-f(u(x))u'(x)$.
MVT for integrals: f cont. $\Rightarrow\exists\xi\in[a,b]:\int_a^bf=f(\xi)(b-a)$.

FACT: free simplifications

- f **odd**, symmetric interval $\Rightarrow\int_{-a}^af=0$. f **even** $\Rightarrow=2\int_0^af$.
- $\int_a^b=-\int_b^a$; $\int_a^b=\int_a^c+\int_c^b$; $|\int f|\le\int|f|$.
- Monotonicity: $f\le g\Rightarrow\int f\le\int g$. Standard estimate: $\left|\int_a^bf\right|\le(b-a)\max|f|$.
- Periodic with period T: $\int_a^{a+T}f$ is independent of a.
- $\int_0^{2\pi}\sin^2=\int_0^{2\pi}\cos^2=\pi$; $\int_0^\pi\sin x\,dx=2$.

▶ RECIPE: limit of a sum = integral (Riemann sum trick)

If a limit looks like $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f\left(\frac{k}{n}\right)$ then it equals $\int_0^1 f(x)\,dx$.

1 Factor out $\frac{1}{n}$ (the width) — if it is not there, the trick does not apply.
2 Write the rest as a function of $\frac{k}{n}$ only.
3 Read off f and integrate over $[0, 1]$. General interval: $\frac{b-a}{n} \sum f\left(a + k\frac{b-a}{n}\right) \to \int_a^b f$.

Ex. $\lim \sum_{k=1}^{n} \frac{n}{n^2+k^2} = \lim \frac{1}{n} \sum \frac{1}{1+(k/n)^2} = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$.

Ex. $\lim \frac{1}{n} \sum_{k=1}^{n} \ln \frac{k}{n} = \int_0^1 \ln x\,dx = -1$ (improper but convergent).

▶ RECIPE: integrals with $|\cdot|$, $\max$, or pieces

1 Find the points where the sign / the active branch changes (zeros of the inside).
2 Split the integral there and drop the $|\cdot|$ with the correct sign on each piece.
3 Add up. **!** Never pull $|\cdot|$ out of the integral.

Ex. $\int_0^2 |x-1|\,dx = \int_0^1 (1-x)\,dx + \int_1^2 (x-1)\,dx = \frac{1}{2} + \frac{1}{2} = 1$.

6.7 Riemann integrability (Riemann-integrierbar)

FACT: criteria (and the false friends)

Definition (Darboux): $\sup_P L(f, P) = \inf_P U(f, P)$; equivalently $\forall\varepsilon \exists P$: $U(f, P) - L(f, P) < \varepsilon$.

- **TRUE:** continuous on $[a, b]$ ⇒ integrable. Monotone on $[a, b]$ ⇒ integrable. Bounded with *finitely many* (even countably many) discontinuities ⇒ integrable. Differentiable ⇒ continuous ⇒ integrable.
- **TRUE (Lebesgue criterion):** bounded + discontinuities form a null set ⟺ integrable.
- **FALSE:** "must be continuous everywhere" — jumps are fine.
- **FALSE:** "integrable ⇒ differentiable" — a step function is integrable, not even continuous.
- **FALSE:** "finitely many function values ⇒ integrable" — Dirichlet $1_{\mathbb{Q}}$ takes only the values $0, 1$ and is *not* integrable.
- **!** integrability requires **bounded** f (otherwise it is an improper integral).

6.8 Improper integrals (uneigentliche Integrale)

▶ RECIPE: does $\int_a^\infty f$ or $\int_a^b f$ (pole) converge?

1 **Locate every problem point:** $\pm\infty$ and every point where f is unbounded. **!** Split the integral so each piece has *exactly one* problem; *all* pieces must converge.
2 Write it as a limit: $\int_a^\infty f = \lim_{R\to\infty} \int_a^R f$, $\int_a^b f = \lim_{\varepsilon\to 0^+} \int_a^{b-\varepsilon} f$.
3 **Benchmarks:** $\int_1^\infty \frac{dx}{x^p}$ conv. ⟺ $p > 1$; $\int_0^1 \frac{dx}{x^p}$ conv. ⟺ $p < 1$. (Remember: at ∞ you need *fast* decay, at 0 you may only be *mildly* singular.) $\int_2^\infty \frac{dx}{x \ln^p x}$ conv. ⟺ $p > 1$. $\int_0^\infty e^{-\lambda x}\,dx = \frac{1}{\lambda}$ $(\lambda > 0)$.
4 **Comparison:** $0 \le f \le g$, $\int g < \infty \Rightarrow \int f < \infty$; $f \ge g \ge 0$, $\int g = \infty \Rightarrow \int f = \infty$.
5 **Limit comparison:** $\frac{f(x)}{g(x)} \to c \in (0, \infty)$ ⇒ both behave the same. Replace f by its leading behaviour at the problem point.
6 **Sign changes:** check absolute convergence first; if that fails, oscillating integrals like $\int_1^\infty \frac{\sin x}{x}$ still converge (Dirichlet/parts) — *conditionally*.

EXAMPLE: which of these converge?

- $\int_0^\infty \frac{\sin t}{t^3}\,dt$: at $t \to 0$, $\frac{\sin t}{t^3} \sim \frac{1}{t^2}$ ⇒ **diverges** (problem is at 0, not at ∞!).
- $\int_0^\infty \frac{dt}{t^2+t^4}$: at $t \to 0$, $\sim \frac{1}{t^2}$ ⇒ **diverges**.
- $\int_1^\infty \frac{\cos^2 t}{t}\,dt$: $\cos^2 t = \frac{1+\cos 2t}{2}$ ⇒ contains $\int \frac{1}{2t}$ ⇒ **diverges**.
- $\int_1^\infty \frac{\cos t}{t}\,dt$: parts with $u = \frac{1}{t}$ ⇒ **converges** (conditionally).

Moral: always inspect *both* ends before answering.

7 Differential equations (ODE)

7.1 Decision tree: which method?

▶ RECIPE: classify the ODE, then jump

1 $y' = f(x)g(y)$ (factorises) ⇒ **separation**, §7.2.
2 $y' + p(x)y = q(x)$ (linear, 1st order) ⇒ **integrating factor**, §7.3.
3 $y' + p(x)y = q(x)y^\alpha$ ⇒ **Bernoulli**: $z = y^{1-\alpha}$, §7.4.
4 $y' = f\left(\frac{y}{x}\right)$ ⇒ substitution $u = \frac{y}{x}$, i.e. $y = ux$, $y' = u'x + u$ ⇒ separable.
5 $y' = f(ax + by + c)$ ⇒ substitution $u = ax + by + c$ ⇒ separable.
6 $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0 y = g(x)$, **constant** a_i ⇒ **characteristic polynomial** §7.5 + particular solution §7.6.
7 $y' = f(y)$ **only (autonomous)** ⇒ separable; constant solutions = zeros of f (check them separately!).

Order of work, always: general solution ⇒ then plug in initial/boundary conditions ⇒ then determine the constants.

7.2 Separation of variables (Trennung der Variablen)

▶ RECIPE: $y' = f(x)g(y)$

1 **! First** find all y_0 with $g(y_0) = 0$: $y \equiv y_0$ are **constant solutions** (often forgotten, costs points).
2 For $g(y) \neq 0$: $\dfrac{dy}{g(y)} = f(x)\,dx$ ⇒ integrate *both* sides, one constant C on the right.
3 Solve for y (keep $\pm$ / $|\cdot|$ carefully; absorb into C: $e^C =: \tilde{C} > 0$, sign gives $\tilde{C} \in \mathbb{R}$).
4 Insert the initial condition, state the interval of existence.

Ex. $y' = xy$: $\frac{dy}{y} = x\,dx \Rightarrow \ln|y| = \frac{x^2}{2} + C \Rightarrow y = \tilde{C}e^{x^2/2}$ (incl. $y \equiv 0$ for $\tilde{C} = 0$).

7.3 First order linear $y' + p(x)y = q(x)$

▶ RECIPE: integrating factor (fastest, always works)

1 Bring to the normal form $y' + p(x)y = q(x)$ (coefficient of y' must be 1!).
2 $\mu(x) := e^{\int p(x)\,dx}$ (no $+C$ needed).
3 $(\mu y)' = \mu q \Rightarrow y = \dfrac{1}{\mu(x)}\left(\displaystyle\int \mu(x)q(x)\,dx + C\right)$.
4 Insert the initial condition last.

Same thing as "variation of constants": $y_h = Ce^{-\int p}$, then let $C = C(x)$, insert, get $C'(x) = qe^{\int p}$, integrate.

EXAMPLE: $y' - \frac{2}{x}y = x^3$

$p = -\frac{2}{x}$, $\mu = e^{-2\ln|x|} = x^{-2}$. $(x^{-2}y)' = x \Rightarrow x^{-2}y = \frac{x^2}{2} + C \Rightarrow y = \frac{x^4}{2} + Cx^2$.

7.4 Bernoulli / substitutions

▶ RECIPE: $y' + p(x)y = q(x)y^\alpha$, $\alpha \neq 0, 1$

1 Divide by y^α: $y^{-\alpha}y' + p\,y^{1-\alpha} = q$.
2 Set $z := y^{1-\alpha} \Rightarrow z' = (1-\alpha)y^{-\alpha}y'$.
3 ⇒ **linear** ODE $z' + (1-\alpha)p\,z = (1-\alpha)q$ ⇒ §7.3.
4 Back-substitute $y = z^{1/(1-\alpha)}$; check the lost solution $y \equiv 0$ separately.

7.5 Linear, constant coefficients — homogeneous part

▶ RECIPE: $y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0 y = 0$

1 Ansatz $y = e^{\lambda x}$ ⇒ **characteristic polynomial** $\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0 = 0$.
2 Find all roots (guess integer roots, then polynomial division).
3 Build the fundamental system:
   - simple real root $\lambda \Rightarrow e^{\lambda x}$
   - real root of multiplicity $m \Rightarrow e^{\lambda x}, xe^{\lambda x}, \ldots, x^{m-1}e^{\lambda x}$
   - complex pair $\lambda = \alpha \pm i\beta \Rightarrow e^{\alpha x}\cos(\beta x)$, $e^{\alpha x}\sin(\beta x)$ (multiplicity m: multiply by $1, x, \ldots, x^{m-1}$)
4 $y_h = C_1 y_1 + \cdots + C_n y_n$ — exactly n free constants.

2nd order quick table $(y'' + by' + cy = 0$, $D = b^2 - 4c)$: $D > 0$: $C_1e^{\lambda_1 x} + C_2e^{\lambda_2 x}$; $D = 0$: $(C_1 + C_2x)e^{\lambda x}$; $D < 0$: $e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x)$.

7.6 Particular solution: undetermined coefficients (Ansatz vom Typ der rechten Seite)

▶ RECIPE: $Ly = g(x)$ — and the resonance rule

1 Pick the ansatz according to g:
   - g = polynomial of degree $k \Rightarrow y_p = A_kx^k + \cdots + A_0$ (full polynomial!)
   - $g = Ce^{\mu x} \Rightarrow y_p = Ae^{\mu x}$
   - $g = C\cos(\omega x)$ or $C\sin(\omega x) \Rightarrow y_p = A\cos(\omega x) + B\sin(\omega x)$ (*both* terms)
   - products ⇒ product of the ansätze, e.g. $xe^{2x} \Rightarrow (Ax + B)e^{2x}$
   - sum $g = g_1 + g_2$ ⇒ solve separately and add (superposition)
2 **! RESONANCE CHECK:** is the exponent μ (resp. $\pm i\omega$) a root of the characteristic polynomial with multiplicity m? ⇒ **multiply the whole ansatz by** x^m. Forgetting this makes the ansatz collapse to $0 = g$.
3 Differentiate the ansatz, insert into the ODE, compare coefficients, solve for $A, B, \ldots$
4 **General solution** $= y_h + y_p$, then impose the initial conditions on the *sum*.

EXAMPLE: $y'' - y = e^x$ (textbook resonance case)

1) $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow y_h = C_1e^x + C_2e^{-x}$.
2) $g = e^x$, $\mu = 1$ *is* a simple root ⇒ resonance, $m = 1$ ⇒ ansatz $y_p = Axe^x$.
3) $y_p' = A(1 + x)e^x$, $y_p'' = A(2 + x)e^x \Rightarrow y_p'' - y_p = 2Ae^x \overset{!}{=} e^x \Rightarrow A = \frac{1}{2}$.
4) $\boxed{y(x) = C_1e^x + C_2e^{-x} + \frac{1}{2}xe^x}$

FACT: variation of constants (when the ansatz method fails, e.g. $g = \frac{1}{x}$, $\tan x$)

2nd order, $y_h = C_1y_1 + C_2y_2$: set $y_p = c_1(x)y_1 + c_2(x)y_2$ with $c_1' = -\dfrac{y_2 g}{W}$, $c_2' = \dfrac{y_1 g}{W}$, Wronskian $W = y_1y_2' - y_1'y_2$; integrate.

EXAMPLE: three more standard ODEs

(a) **Separable + IVP.** $y' = y^2\cos x$, $y(0) = 1$: constant solution $y \equiv 0$; else $-\frac{1}{y} = \sin x + C \Rightarrow y = \frac{-1}{\sin x + C}$; $y(0) = 1 \Rightarrow C = -1 \Rightarrow y = \frac{1}{1-\sin x}$.
(b) **Bernoulli.** $y' + y = xy^2$ $(\alpha = 2)$: $z = y^{-1} \Rightarrow z' - z = -x \Rightarrow \mu = e^{-x} \Rightarrow z = x + 1 + Ce^x \Rightarrow y = \frac{1}{x+1+Ce^x}$.
(c) **Complex roots + resonant** $\cos$. $y'' + y = \cos x$: $\lambda = \pm i \Rightarrow y_h = C_1\cos x + C_2\sin x$. $\pm i$ *is* a root ⇒ resonance ⇒ ansatz $y_p = x(A\cos x + B\sin x) \Rightarrow A = 0$, $B = \frac{1}{2} \Rightarrow y = C_1\cos x + C_2\sin x + \frac{x}{2}\sin x$.

7.7 Initial / boundary conditions, existence

FACT: IVP vs. BVP, Picard–Lindelöf

IVP *(Anfangswertproblem)*: all conditions at one point x_0: $y(x_0), y'(x_0), \ldots$ — n conditions for order n. **BVP** *(Randwertproblem)*: conditions at two points a, b — may have 0, 1 or ∞ many solutions.

Picard–Lindelöf: $y' = f(x, y)$, $y(x_0) = y_0$ with f continuous and *Lipschitz in y* (e.g. $\partial_y f$ continuous & bounded locally) ⇒ **unique** solution on a neighbourhood of x_0. **!** Continuity of f alone gives existence (Peano) but *not* uniqueness: $y' = \sqrt{|y|}$, $y(0) = 0$ has $y \equiv 0$ and $y = \frac{x^2}{4}$. Linear ODEs with continuous coefficients: unique solution on the whole interval.

8 Standard proofs from the lecture (Vorlesungsbeweise)

8.1 How to write a proof for points

▶ RECIPE: generic scheme

1 State what is given and what is to be shown ("Zu zeigen: ...").
2 Reduce w.l.o.g. (*oBdA*) — e.g. replace f by $-f$ to fix a sign.
3 Build the auxiliary object (a set, a helper function, a subsequence). *This is where most points sit.*
4 Justify existence (completeness / Bolzano–Weierstrass / Weierstrass max).
5 Do the ε-argument or the contradiction.
6 Close with the statement + □.

8.2 Intermediate value theorem (Zwischenwertsatz)

FACT: $f \in C([a,b])$, c between $f(a), f(b) \Rightarrow \exists x : f(x) = c$

1 oBdA $f(a) \le c \le f(b)$ (else consider $-f$).
2 Define $X := \{x \in [a,b] \mid f(x) \le c\}$.
3 $X \neq \emptyset$ ($a \in X$) and $X \subseteq [a,b]$ bounded $\Rightarrow x^* := \sup X$ exists (completeness).
4 Choose $x_n \in X \cap [x^* - \frac{1}{n}, x^*] \Rightarrow x_n \to x^*$ and $f(x_n) \le c$.
5 Continuity (sequential) $\Rightarrow f(x^*) = \lim f(x_n) \le c$.
6 Assume $f(x^*) < c$. Then $x^* < b$ (since $c \le f(b)$). Put $\varepsilon := c - f(x^*) > 0$; continuity gives $\delta > 0$ with $|y - x^*| < \delta \Rightarrow |f(y) - f(x^*)| < \varepsilon$.
7 Hence $f(y) < f(x^*) + \varepsilon = c$ for those y, i.e. $(x^*, x^* + \delta) \cap [a,b] \subseteq X$.
8 Contradiction to $x^* = \sup X \Rightarrow f(x^*) = c$. $\square$

8.3 Cauchy sequence converges (Vollständigkeit von $\mathbb{R}$)

FACT: (a_n) Cauchy in $\mathbb{R} \Rightarrow$ convergent

1 Cauchy $\Rightarrow$ **bounded** (take $\varepsilon = 1$, N; then $|a_n| \le \max\{|a_1|, \dots, |a_N|, |a_N| + 1\}$).
2 Bolzano–Weierstrass $\Rightarrow$ a convergent subsequence $a_{n_k} \to L$.
3 Cauchy: $\exists N$ with $|a_n - a_m| < \frac{\varepsilon}{2}$ for all $n, m \ge N$.
4 Subsequence: $\exists N_1$ with $|a_{n_k} - L| < \frac{\varepsilon}{2}$ for all $k \ge N_1$.
5 $N^* := \max\{N, N_1\}$; using $n_k \ge k$ pick one index $n_k \ge N^*$.
6 Triangle inequality: $|a_n - L| \le |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ for all $n \ge N^* \Rightarrow a_n \to L$. $\square$

8.4 Rolle and the mean value theorem

FACT: Rolle: $f \in C[a,b]$, diff. on (a,b), $f(a) = f(b) \Rightarrow \exists \xi : f'(\xi) = 0$

1 f continuous on the compact $[a,b] \Rightarrow$ attains max and min (Weierstrass).
2 If both are attained at the boundary then f is constant $\Rightarrow f' \equiv 0$, done.
3 Otherwise an extremum lies in the interior at some $\xi \Rightarrow$ Fermat $\Rightarrow f'(\xi) = 0$. $\square$

FACT: MVT (Mittelwertsatz): $\exists \xi \in (a,b) : f'(\xi) = \frac{f(b)-f(a)}{b-a}$

1 Auxiliary function $g(x) := f(x) - \dfrac{f(b) - f(a)}{b - a}(x - a)$.
2 g is continuous on $[a,b]$, differentiable on (a,b), and $g(a) = f(a) = g(b)$.
3 Rolle $\Rightarrow \exists \xi \in (a,b)$ with $0 = g'(\xi) = f'(\xi) - \frac{f(b)-f(a)}{b-a}$. $\square$

8.5 Further one-liners you may be asked

FACT: differentiable $\Rightarrow$ continuous

$f(x) - f(x_0) = \frac{f(x)-f(x_0)}{x-x_0}(x - x_0) \to f'(x_0) \cdot 0 = 0$. $\square$

FACT: convergent $\Rightarrow$ bounded

Take $\varepsilon = 1$, get N with $|a_n - a| < 1$ for $n \ge N \Rightarrow |a_n| \le |a| + 1$; the finitely many $a_1, \dots, a_{N-1}$ are bounded by their maximum. $\square$

FACT: limit is unique

Assume $a_n \to a$ and $a_n \to b$, $a \neq b$. Take $\varepsilon = \frac{|a-b|}{2}$; for large n: $|a - b| \le |a - a_n| + |a_n - b| < |a - b|$, contradiction. $\square$

FACT: monotone + bounded $\Rightarrow$ convergent

(a_n) increasing, bounded $\Rightarrow s := \sup\{a_n\}$ exists. Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound $\Rightarrow \exists N : a_N > s - \varepsilon$; monotonicity $\Rightarrow s - \varepsilon < a_N \le a_n \le s$ for $n \ge N \Rightarrow |a_n - s| < \varepsilon$. $\square$

FACT: Bolzano–Weierstrass

Bisection: $(a_n) \subseteq [A, B]$. Halve the interval and keep a half containing infinitely many terms; repeat $\Rightarrow$ nested intervals of length $\frac{B-A}{2^k}$. Pick n_k increasing with a_{n_k} in the k-th interval $\Rightarrow$ the nested intervals shrink to a point L and $a_{n_k} \to L$. $\square$

FACT: uniform limit of continuous functions is continuous

$\frac{\varepsilon}{3}$-argument: $|f(x) - f(x_0)| \le |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)|$. Choose N with $\|f_N - f\|_\infty < \frac{\varepsilon}{3}$ (uniformity: same N for all x!), then δ from continuity of f_N at x_0. $\square$

FACT: absolute convergence $\Rightarrow$ convergence

Cauchy criterion: $\left|\sum_{k=n}^m a_k\right| \le \sum_{k=n}^m |a_k| < \varepsilon$ for $m \ge n \ge N$; the partial sums are Cauchy $\Rightarrow$ converge. $\square$

FACT: Fermat

x_0 interior local max $\Rightarrow$ for $h > 0$ small $\frac{f(x_0+h)-f(x_0)}{h} \le 0$ and for $h < 0$ it is ≥ 0; both one-sided limits equal $f'(x_0) \Rightarrow f'(x_0) = 0$. $\square$

FACT: squeeze theorem

$b_n \le a_n \le c_n$, $b_n, c_n \to L$. Given ε: for large n, $L - \varepsilon < b_n \le a_n \le c_n < L + \varepsilon \Rightarrow |a_n - L| < \varepsilon$. $\square$

FACT: limit laws (sum, product)

$|(a_n + b_n) - (a + b)| \le |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$. Product: $|a_n b_n - ab| \le |a_n||b_n - b| + |b||a_n - a|$; (a_n) is bounded by C (convergent $\Rightarrow$ bounded) $\Rightarrow \le C|b_n - b| + |b||a_n - a| < \varepsilon$. $\square$

FACT: product rule

$\frac{(fg)(x)-(fg)(x_0)}{x-x_0} = f(x)\frac{g(x)-g(x_0)}{x-x_0} + g(x_0)\frac{f(x)-f(x_0)}{x-x_0}$; let $x \to x_0$ and use $f(x) \to f(x_0)$ (continuity). $\square$

FACT: chain rule

Put $\varphi(y) := \frac{g(y)-g(y_0)}{y-y_0}$ for $y \neq y_0$ and $\varphi(y_0) := g'(y_0)$; φ is continuous at y_0 and $g(y) - g(y_0) = \varphi(y)(y - y_0)$. Insert $y = f(x)$, $y_0 = f(x_0)$, divide by $x - x_0$, let $x \to x_0$. $\square$

FACT: derivative of the inverse

f^{-1} continuous, $f'(x_0) \neq 0$: $\frac{f^{-1}(y)-f^{-1}(y_0)}{y-y_0} = \frac{x-x_0}{f(x)-f(x_0)} \to \frac{1}{f'(x_0)}$ as $y \to y_0$ (then $x \to x_0$). $\square$

FACT: extreme value theorem (Weierstrass)

f cont. on $[a,b]$. **(i) bounded:** else $\exists x_n$ with $|f(x_n)| \ge n$; Bolzano–Weierstrass $\Rightarrow x_{n_k} \to x^* \in [a,b]$, continuity $\Rightarrow f(x_{n_k}) \to f(x^*)$ finite — contradiction. **(ii) attained:** $S := \sup f$ exists; pick x_n with $f(x_n) \to S$; a convergent subsequence $x_{n_k} \to x^*$ gives $f(x^*) = S$. $\square$

FACT: continuous on a compact interval $\Rightarrow$ uniformly continuous (Heine)

Suppose not: $\exists \varepsilon_0 > 0$ and x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| \ge \varepsilon_0$. Bolzano–Weierstrass $\Rightarrow x_{n_k} \to x^*$, hence also $y_{n_k} \to x^*$. Continuity $\Rightarrow |f(x_{n_k}) - f(y_{n_k})| \to 0$ — contradiction. $\square$

FACT: fundamental theorem of calculus

$F(x) = \int_a^x f$. Then $\frac{F(x+h)-F(x)}{h} = \frac{1}{h}\int_x^{x+h} f = f(\xi_h)$ for some ξ_h between x and $x + h$ (MVT for integrals). $h \to 0 \Rightarrow \xi_h \to x \Rightarrow$ continuity $\Rightarrow F'(x) = f(x)$. $\square$

FACT: geometric series

$s_N = \sum_{k=0}^N q^k$; $s_N - q s_N = 1 - q^{N+1} \Rightarrow s_N = \frac{1-q^{N+1}}{1-q}$. $|q| < 1 \Rightarrow q^{N+1} \to 0 \Rightarrow s_N \to \frac{1}{1-q}$. $\square$

FACT: root test

$\limsup \sqrt[n]{|a_n|} = q < 1$: pick $q < r < 1$; for $n \ge N$, $\sqrt[n]{|a_n|} \le r \Rightarrow |a_n| \le r^n \Rightarrow$ comparison with the convergent geometric series. $q > 1 \Rightarrow |a_n| \ge 1$ infinitely often $\Rightarrow a_n \not\to 0$. $\square$

FACT: Cauchy–Schwarz

For all t: $0 \le \|x + ty\|^2 = \|x\|^2 + 2t\langle x, y\rangle + t^2\|y\|^2$. A quadratic in t that is ≥ 0 has discriminant ≤ 0: $4\langle x, y\rangle^2 - 4\|x\|^2\|y\|^2 \le 0$. $\square$

FACT: a convergent sequence has exactly one accumulation point

$a_n \to a$: every subsequence also $\to a$ (its indices $n_k \ge k$), so a is the only possible subsequential limit. $\square$

FACT: $\left(1 + \frac{1}{n}\right)^n$ converges

Increasing by AM–GM (or the binomial theorem term-by-term) and bounded above by $\sum_{k \ge 0} \frac{1}{k!} < 3 \Rightarrow$ monotone + bounded $\Rightarrow$ converges; the limit is called e. $\square$

FACT: monotone functions are Riemann integrable

f increasing on $[a,b]$, equidistant partition with mesh $\frac{b-a}{n}$: $U(f,P) - L(f,P) = \frac{b-a}{n}\sum_i (f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) \to 0$ (telescoping) $\Rightarrow$ integrable. $\square$

FACT: convergent $\Rightarrow$ Cauchy

$|a_n - a_m| \le |a_n - a| + |a - a_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$ for $n, m \ge N$. $\square$

FACT: Archimedean property

If $\mathbb{N}$ were bounded above, $s := \sup \mathbb{N}$ would exist; $s - 1$ is not an upper bound $\Rightarrow \exists n > s - 1 \Rightarrow n + 1 > s$, contradiction. $\Rightarrow \forall x\ \exists n > x$, hence $\frac{1}{n} \to 0$. $\square$

FACT: characterisation of the supremum

$s = \sup M \iff$ (i) $x \le s\ \forall x \in M$ and (ii) $\forall \varepsilon > 0\ \exists x \in M : x > s - \varepsilon$. "$\Rightarrow$": if (ii) failed, $s - \varepsilon$ would be a smaller upper bound. "$\Leftarrow$": (i) makes s an upper bound, (ii) makes it the least. $\square$

FACT: comparison test

$0 \le a_n \le b_n$, $\sum b_n < \infty$: the partial sums of $\sum a_n$ are increasing and bounded by $\sum b_n \Rightarrow$ monotone + bounded $\Rightarrow$ converge. $\square$

FACT: Leibniz criterion

$b_n \downarrow 0$: (s_{2n}) is increasing, (s_{2n+1}) decreasing, $s_{2n} \le s_{2n+1}$, and $s_{2n+1} - s_{2n} = b_{2n+1} \to 0 \Rightarrow$ nested intervals $\Rightarrow$ both converge to the same s. $\square$

FACT: Cauchy–Hadamard

Apply the root test to $\sum a_n(x - x_0)^n$: $\limsup \sqrt[n]{|a_n||x - x_0|^n} = |x - x_0| \limsup \sqrt[n]{|a_n|} = \frac{|x-x_0|}{R}$, which is < 1 exactly for $|x - x_0| < R$. $\square$

FACT: uniform convergence allows swapping $\lim$ and $\int$

$\left|\int_a^b f_n - \int_a^b f\right| \le \int_a^b |f_n - f| \le (b - a)\|f_n - f\|_\infty \to 0$. $\square$

FACT: l'Hospital (idea)

$f(x_0) = g(x_0) = 0$: Cauchy MVT gives $\frac{f(x)}{g(x)} = \frac{f(x)-f(x_0)}{g(x)-g(x_0)} = \frac{f'(\xi_x)}{g'(\xi_x)}$ with ξ_x between x_0 and x; $x \to x_0$ forces $\xi_x \to x_0$. $\square$

FACT: convex $\iff f'' \ge 0$ (differentiable case)

f convex $\iff f'$ monotonically increasing (chord slopes increase) $\iff f'' \ge 0$ (monotonicity criterion). $\square$

FACT: contraction / uniqueness of a fixed point

$|f(x) - f(y)| \le q|x - y|$, $q < 1$. Two fixed points $\Rightarrow |x^* - y^*| \le q|x^* - y^*| \Rightarrow (1 - q)|x^* - y^*| \le 0 \Rightarrow x^* = y^*$. Existence: $x_{n+1} = f(x_n)$ is Cauchy since $|x_{n+1} - x_n| \le q^n|x_1 - x_0|$. $\square$

9 Lookup tables

TRAP: HOW TO USE THE DERIVATIVE / ANTIDERIVATIVE TABLE

Read a row as $\frac{d}{dx}F(x) = f(x)$ and $\int f(x)\,dx = F(x) + C$. **The table is written for the bare argument x. (1) Inner function is linear $ax+b$:** $\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$ — divide by the inner derivative a. **(2) Inner function is anything else:** you may *not* patch with a factor; substitute $u = g(x)$, $dx = \frac{du}{g'(x)}$, and only proceed if the leftover $g'(x)$ cancels. **(3) Constant prefactors** stay outside: $\int c\,f = c\int f$. Do the substitution factor and the prefactor as *separate* steps and write both down. **(4) Always differentiate your answer at the end** — the chain rule factor must reappear. Example: $\int \frac{1}{2\cos^2(x/2)}\,dx = \tan\frac{x}{2} + C$ (*not* $2\tan\frac{x}{2}$), because $dx = 2\,du$ cancels the $\frac{1}{2}$.

Antiderivative ← Function → Derivative

$F(x) = \int f$	$f(x)$	$f'(x)$
$\frac{x^{n+1}}{n+1}$ $(n \neq -1)$	x^n	nx^{n-1}
$\ln\lvert x\rvert$	$\frac{1}{x}$	$-\frac{1}{x^2}$
$\frac{2}{3}x^{3/2}$	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
$\frac{n}{n+1}x^{(n+1)/n}$	$\sqrt[n]{x}$	$\frac{1}{n}x^{1/n-1}$
$\frac{1}{a(n+1)}(ax+b)^{n+1}$	$(ax+b)^n$	$an(ax+b)^{n-1}$
e^x	e^x	e^x
$\frac{1}{a}e^{ax}$	e^{ax}	ae^{ax}
$\frac{a^x}{\ln a}$	a^x	$a^x\ln a$
$x(\ln\lvert x\rvert - 1)$	$\ln\lvert x\rvert$	$\frac{1}{x}$
$\frac{x}{\ln a}(\ln\lvert x\rvert - 1)$	$\log_a\lvert x\rvert$	$\frac{1}{x\ln a}$
$-\cos x$	$\sin x$	$\cos x$
$\sin x$	$\cos x$	$-\sin x$
$-\ln\lvert\cos x\rvert$	$\tan x$	$\frac{1}{\cos^2 x} = 1 + \tan^2 x$
$\ln\lvert\sin x\rvert$	$\cot x$	$-\frac{1}{\sin^2 x}$
$\ln\left\lvert\tan\frac{x}{2}\right\rvert$	$\frac{1}{\sin x}$	$-\frac{\cos x}{\sin^2 x}$
$\ln\left\lvert\tan\left(\frac{x}{2}+\frac{\pi}{4}\right)\right\rvert$	$\frac{1}{\cos x}$	$\frac{\sin x}{\cos^2 x}$
$\frac{x}{2} - \frac{\sin 2x}{4}$	$\sin^2 x$	$\sin 2x$
$\frac{x}{2} + \frac{\sin 2x}{4}$	$\cos^2 x$	$-\sin 2x$
$x\arcsin x + \sqrt{1-x^2}$	$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
$x\arccos x - \sqrt{1-x^2}$	$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
$x\arctan x - \frac{1}{2}\ln(1+x^2)$	$\arctan x$	$\frac{1}{1+x^2}$
$\cosh x$	$\sinh x$	$\cosh x$
$\sinh x$	$\cosh x$	$\sinh x$
$\ln\cosh x$	$\tanh x$	$\frac{1}{\cosh^2 x}$
$\operatorname{arsinh} x$	$\frac{1}{\sqrt{1+x^2}}$	$\frac{-x}{(1+x^2)^{3/2}}$

Antiderivatives you will not guess

$f(x)$	$F(x) = \int f\,dx$
$\frac{1}{ax+b}$	$\frac{1}{a}\ln\lvert ax+b\rvert$
$\frac{f'(x)}{f(x)}$	$\ln\lvert f(x)\rvert$
$\frac{1}{a^2+x^2}$	$\frac{1}{a}\arctan\frac{x}{a}$
$\frac{1}{a^2-x^2}$	$\frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert$
$\frac{1}{\sqrt{a^2-x^2}}$	$\arcsin\frac{x}{\lvert a\rvert}$
$\frac{1}{\sqrt{x^2\pm a^2}}$	$\ln\left\lvert x + \sqrt{x^2 \pm a^2}\right\rvert$
$\sqrt{a^2-x^2}$	$\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{\lvert a\rvert}$
$\sqrt{a^2+x^2}$	$\frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2}\ln\left\lvert x + \sqrt{a^2+x^2}\right\rvert$
$\sqrt{x^2-a^2}$	$\frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left\lvert x + \sqrt{x^2-a^2}\right\rvert$
$\ln x$	$x\ln x - x$
xe^{ax}	$\frac{e^{ax}}{a^2}(ax-1)$
$e^{ax}\sin(bx)$	$\frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$
$e^{ax}\cos(bx)$	$\frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$
$\frac{1}{1+\cos x}$	$\tan\frac{x}{2}$
$\frac{1}{1-\cos x}$	$-\cot\frac{x}{2}$
$\frac{1}{\sin x\cos x}$	$\ln\lvert\tan x\rvert$
x^x	no elementary form ($\frac{d}{dx}x^x = x^x(1+\ln x)$)

Growth hierarchy ($x\to\infty$, $\alpha > 0$, $b > 1$): $\ln x \ll x^\alpha \ll b^x \ll x! \ll x^x$ — every ratio of a left to a right term $\to 0$. Use it to kill limits instantly.

Standard limits

- $\lim_{x\to 0}\frac{\sin x}{x} = 1$
- $\lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}$
- $\lim_{x\to 0}\frac{\tan x}{x} = 1$
- $\lim_{x\to 0}\frac{\arctan x}{x} = 1$
- $\lim_{x\to 0}\frac{e^x-1}{x} = 1$
- $\lim_{x\to 0}\frac{a^x-1}{x} = \ln a$
- $\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1$
- $\lim_{x\to 1}\frac{\ln x}{x-1} = 1$
- $\lim_{x\to 0^+} x\ln x = 0$
- $\lim_{x\to 0^+} x^x = 1$
- $\lim_{x\to\infty}\frac{\ln x}{x^\alpha} = 0$
- $\lim_{x\to\infty}\frac{x^m}{e^x} = 0$
- $\lim_{n\to\infty}\sqrt[n]{n} = 1$
- $\lim_{n\to\infty}\sqrt[n]{a} = 1$ $(a > 0)$
- $\lim_{n\to\infty}\frac{a^n}{n!} = 0$
- $\lim_{n\to\infty}\sqrt[n]{n!} = \infty$
- $\lim_{x\to\pm\infty}(1+\frac{k}{x})^{mx} = e^{km}$
- $\lim_{x\to 0}(1+x)^{1/x} = e$
- $\lim_{x\to\infty} x^\alpha q^x = 0$ $(0 \le q < 1)$
- $\lim_{x\to\infty}\arctan x = \frac{\pi}{2}$

Trigonometry

- $\sin^2 x + \cos^2 x = 1$; $1 + \tan^2 x = \frac{1}{\cos^2 x}$
- $\sin(x\pm y) = \sin x\cos y \pm \cos x\sin y$
- $\cos(x\pm y) = \cos x\cos y \mp \sin x\sin y$
- $\sin 2x = 2\sin x\cos x$; $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$
- $\sin^2 x = \frac{1-\cos 2x}{2}$; $\cos^2 x = \frac{1+\cos 2x}{2}$
- $1 + \cos x = 2\cos^2\frac{x}{2}$; $1 - \cos x = 2\sin^2\frac{x}{2}$
- $\tan\frac{x}{2} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$
- $\sin(-x) = -\sin x$ (odd), $\cos(-x) = \cos x$ (even)
- $\sin(\pi - x) = \sin x$, $\cos(\pi - x) = -\cos x$
- $\sin(x + \frac{\pi}{2}) = \cos x$, $\cos(x + \frac{\pi}{2}) = -\sin x$
- $\arcsin : [-1,1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\arccos : [-1,1] \to [0,\pi]$, $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$

deg	rad	sin	cos	tan
0	0	0	1	0
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90	$\frac{\pi}{2}$	1	0	–
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180	π	0	-1	0

Quad.	sin	cos	tan
I	+	+	+
II	+	–	–
III	–	–	+
IV	–	+	–

Logs, powers, hyperbolics

- $\log_a x = \frac{\ln x}{\ln a}$; $a^x = e^{x\ln a}$; $u^v = e^{v\ln u}$
- $\ln(xy) = \ln x + \ln y$; $\ln\frac{x}{y} = \ln x - \ln y$; $\ln(x^y) = y\ln x$
- $\ln 1 = 0$, $\ln e = 1$; $\ln x$ defined only for $x > 0$
- $\sinh x = \frac{e^x - e^{-x}}{2}$, $\cosh x = \frac{e^x + e^{-x}}{2}$, $\cosh^2 - \sinh^2 = 1$
- $\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$, $\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1})$, $\operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$

Algebra survival kit

- $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A}+\sqrt{B}}$ (conjugate trick, kills $\infty - \infty$)
- $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$
- Complete the square: $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$
- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$; $(a+b)^n = \sum_k \binom{n}{k} a^k b^{n-k}$
- $\sum_{k=1}^n k = \frac{n(n+1)}{2}$; $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$; $\sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$
- Guess a root of a polynomial among the divisors of the constant term, then divide.

Taylor series at 0 (with radius)

- $e^x = \sum_{n\ge 0}\frac{x^n}{n!}$, $R = \infty$
- $\sin x = \sum_{n\ge 0}\frac{(-1)^n x^{2n+1}}{(2n+1)!}$, $R = \infty$
- $\cos x = \sum_{n\ge 0}\frac{(-1)^n x^{2n}}{(2n)!}$, $R = \infty$
- $\frac{1}{1-x} = \sum_{n\ge 0} x^n$, $R = 1$

- $\ln(1+x) = \sum_{n\ge 1} \frac{(-1)^{n+1}x^n}{n}$, $R = 1$
- $\arctan x = \sum_{n\ge 0} \frac{(-1)^n x^{2n+1}}{2n+1}$, $R = 1$
- $(1+x)^\alpha = \sum_{n\ge 0} \binom{\alpha}{n} x^n$, $R = 1$
- $\sinh x = \sum \frac{x^{2n+1}}{(2n+1)!}$, $\cosh x = \sum \frac{x^{2n}}{(2n)!}$, $R = \infty$

I need to...	$\to$ use
show a sequence converges without knowing the limit	monotone+bounded, or Cauchy
show a value is attained	IVT / Weierstrass max-min
show a zero or fixed point exists	IVT on $f(x) - x$
show it is unique	strict monotonicity ($h' \neq 0$) or Rolle
classify a critical point	f'' sign; if $f'' = 0$ use sign change of f'
evaluate $\frac{0}{0}$ or $\frac{\infty}{\infty}$	Taylor (preferred) or l'Hospital
handle $0 \cdot \infty$, $\infty - \infty$, 1^∞	rewrite first, then l'Hospital
decide $\int_a^\infty f$ converges	compare with x^{-p}, check *both* ends
integrate a root $\sqrt{a^2 \pm x^2}$	trig substitution
integrate a rational function	partial fractions
integrate rational in $\sin, \cos$	half-angle, else $t = \tan \frac{x}{2}$
solve a forced linear ODE	$y_h + y_p$, check resonance
solve a 1st order nonlinear ODE	separation / $u = y/x$ / Bernoulli / int. factor
show $f_n \rightrightarrows f$	$\sup \lvert f_n - f\rvert \to 0$
disprove $f_n \rightrightarrows f$	one sequence x_n with $\lvert f_n(x_n) - f(x_n)\rvert \ge \varepsilon_0$
swap lim and $\int$	needs *uniform* convergence

Series looks like...	$\to$ take
$a_n \not\to 0$	divergence test, done
$\sum \frac{1}{n^p}$, $\sum q^n$	p-series / geometric
factorials	ratio test
n-th powers	root test
alternating, $b_n \downarrow 0$	Leibniz
looks like a known series	comparison (leading powers)
$a_n = f(n)$, $f \downarrow$	integral test
power series	Cauchy–Hadamard + endpoints

Counterexample arsenal (kill a false MC option in 5 s)

"$a_{n+1} - a_n \to 0 \Rightarrow$ convergent"	$a_n = \sqrt{n}$, $a_n = \sum_{k\le n} \frac{1}{k}$
"bounded $\Rightarrow$ convergent"	$a_n = (-1)^n$
"$a_n \to 0 \Rightarrow \sum a_n$ conv."	$\sum \frac{1}{n}$
"continuous $\Rightarrow$ differentiable"	$\lvert x\rvert$ at 0
"differentiable $\Rightarrow C^1$"	$x^2 \sin \frac{1}{x}$, $f(0) = 0$
"$f'(x_0) = 0 \Rightarrow$ extremum"	x^3 at 0
"$f''(x_0) = 0 \Rightarrow$ inflection"	x^4 at 0
"pointwise $\Rightarrow$ uniform"	x^n on $[0,1]$
"pointwise $\Rightarrow$ limit continuous"	x^n on $[0,1]$
"$f_n \rightrightarrows f \Rightarrow f_n' \to f'$"	$\frac{\sin(nx)}{n}$
"pointwise $\Rightarrow \int f_n \to \int f$"	$n^2 x e^{-nx}$ on $[0,1]$
"finitely many values $\Rightarrow$ integrable"	Dirichlet $1_{\mathbb{Q}}$
"integrable $\Rightarrow$ continuous"	step function
"IVT applies on any interval"	$\frac{1}{x}$ on $[-1,1]$ (not continuous)
"$\sum a_n$, $\sum b_n$ conv. $\Rightarrow \sum a_n b_n$ conv."	$a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$
"continuous on $\mathbb{R} \Rightarrow$ unif. continuous"	x^2, $\sin(x^2)$
"f cont. $\Rightarrow$ attains max"	$\frac{1}{x}$ on $(0,1]$ (not compact)
"$f > 0$, $\int_1^\infty f < \infty \Rightarrow f \to 0$"	thin spikes of height 1
"unique ODE solution always"	$y' = \sqrt{\lvert y\rvert}$, $y(0) = 0$

Negate a definition (exam loves this)

Rule: swap $\forall \leftrightarrow \exists$ from left to right, negate the final inequality ($<$ becomes $\ge$).

- $a_n \not\to a$: $\exists \varepsilon > 0\ \forall N\ \exists n \ge N:\ |a_n - a| \ge \varepsilon$.
- **(a_n) not Cauchy:** $\exists \varepsilon > 0\ \forall N\ \exists m, n \ge N:\ |a_n - a_m| \ge \varepsilon$.
- **f not continuous at x_0:** $\exists \varepsilon > 0\ \forall \delta > 0\ \exists x:\ |x - x_0| < \delta$ and $|f(x) - f(x_0)| \ge \varepsilon$.
- **f not uniformly continuous:** $\exists \varepsilon > 0\ \forall \delta > 0\ \exists x, y:\ |x - y| < \delta$ and $|f(x) - f(y)| \ge \varepsilon$. $\Rightarrow$ in practice: build sequences x_n, y_n with $|x_n - y_n| \to 0$, $|f(x_n) - f(y_n)| \ge \varepsilon$.
- $f_n \not\rightrightarrows f$: $\exists \varepsilon > 0\ \forall N\ \exists n \ge N\ \exists x:\ |f_n(x) - f(x)| \ge \varepsilon$. $\Rightarrow$ in practice: one sequence x_n with $|f_n(x_n) - f(x_n)| \ge \varepsilon$.
- **M unbounded above:** $\forall K\ \exists x \in M:\ x > K$.

Implication map — which way the arrow points

- Sequences: convergent $\Rightarrow$ Cauchy $\Rightarrow$ bounded $\Rightarrow$ has convergent subsequence. In $\mathbb{R}$: Cauchy $\Leftrightarrow$ convergent.
- Monotone + bounded $\Leftrightarrow$ convergent (for monotone sequences).
- Series: abs. conv. $\Rightarrow$ conv. $\Rightarrow a_n \to 0$. None reverses.
- Functions: $C^1 \Rightarrow$ differentiable $\Rightarrow$ continuous $\Rightarrow$ (on compact) uniformly continuous $\Rightarrow$ (bounded) Riemann integrable.
- Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous.
- $f_n \rightrightarrows f \Rightarrow f_n \to f$ pointwise. Reverse only with extra hypotheses (Dini: monotone + compact + f continuous).
- Convex + differentiable $\Rightarrow f'$ increasing $\Rightarrow$ critical point is a global min.

Exam-day checklist

- **Integral asked, answer given?** Differentiate the options instead of integrating.
- **Limit $x \to 0$?** Taylor beats l'Hospital nearly always.
- **Improper integral?** Check *both* endpoints, split first.
- **Separable ODE?** Write down the constant solutions $g(y) = 0$.
- **Forced linear ODE?** Resonance check before the ansatz.
- **"Show it converges"?** Name the theorem you use (monotone+bounded / Cauchy / comparison) — naming earns points.
- **"Justify"?** One sentence per step, plus the hypotheses of the theorem you cite ("f continuous on the *compact* $[a,b]$, therefore ...").
- **Substitution done?** Differentiate the result and check the chain-rule factor.
- **Uniform convergence?** Compute $\sup |f_n - f|$ explicitly; a maximiser x_n is a complete answer.
- **Stuck on a proof?** Write the definitions of everything given and everything to show — that alone is often half the points.
- **Multiple choice, no negative points:** never leave a box empty.

Word list DE $\to$ EN

Folge = sequence • Reihe = series • Grenzwert = limit • Häufungspunkt = accumulation point • beschränkt = bounded • monoton = monotone • Teilfolge = subsequence • stetig = continuous • gleichmässig = uniform(ly) • punktweise = pointwise • differenzierbar = differentiable • Ableitung = derivative • Stammfunktion = antiderivative • Zwischenwertsatz = IVT • Mittelwertsatz = MVT • Umkehrfunktion = inverse • Nullstelle = root/zero • Wendepunkt = inflection point • Krümmung = curvature • Abschätzung = estimate • oBdA = w.l.o.g. • Voraussetzung = hypothesis • Behauptung = claim • Widerspruch = contradiction • Anfangswertproblem = IVP • Randwertproblem = BVP • Trennung der Variablen = separation of variables • uneigentlich = improper • Konvergenzradius = radius of convergence.

EXAMPLE: model solution — "Untersuchen Sie die Funktionenfolge $f_n(x) = \frac{x}{1+nx^2}$ auf $\mathbb{R}$" (9 P)

a) Gleichmässige Stetigkeit. For $n = 0$ we have $f_0(x) = x$, which is Lipschitz with constant 1, hence uniformly continuous. For $n \ge 1$, f_n is differentiable on $\mathbb{R}$ with $f_n'(x) = \frac{1-nx^2}{(1+nx^2)^2}$, so $|f_n'(x)| \le \frac{1+nx^2}{(1+nx^2)^2} = \frac{1}{1+nx^2} \le 1$. By the mean value theorem $|f_n(x) - f_n(y)| \le |x - y|$ for all x, y, so with $\delta := \varepsilon$ the definition of uniform continuity is satisfied. Hence *all* f_n are uniformly continuous.
b) Punktweiser Grenzwert. Fix x. For $x = 0$: $f_n(0) = 0$ for all n. For $x \neq 0$: $|f_n(x)| = \frac{|x|}{1+nx^2} \le \frac{|x|}{nx^2} = \frac{1}{n|x|} \to 0$. Hence $f_n \to f \equiv 0$ pointwise on $\mathbb{R}$.
c) Gleichmässige Konvergenz. We compute $\|f_n - f\|_\infty = \sup_x |f_n(x)|$. $f_n'(x) = 0 \iff x = \pm\frac{1}{\sqrt{n}}$, and $f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1}{2\sqrt{n}}$, $f_n(x) \to 0$ for $|x| \to \infty$. Therefore $\|f_n\|_\infty = \frac{1}{2\sqrt{n}} \to 0$, i.e. $f_n \rightrightarrows 0$ **uniformly on $\mathbb{R}$**. $\square$

EXAMPLE: model solution — "$f(x) = \frac{\sin x}{1+x^2}$: Fixpunkt und Rekursion" (9 P)

a) Eindeutiger Fixpunkt und $|f(x)| < |x|$ für $x \neq 0$. $f(0) = 0$, so 0 is a fixed point. For $x \neq 0$ we use the strict inequality $|\sin x| < |x|$ (valid for all $x \neq 0$) and $1 + x^2 \ge 1$:

$$|f(x)| = \frac{|\sin x|}{1+x^2} \le |\sin x| < |x|.$$

If x^* were a further fixed point, then $|x^*| = |f(x^*)| < |x^*|$ — contradiction. So $x^* = 0$ is the **unique** fixed point.
b) Konvergenzverhalten von $x_{n+1} = f(x_n)$, $x_0 \in (0,1)$. *Positivity:* for $x \in (0,1)$ we have $\sin x > 0$ and $1 + x^2 > 0$, so $f(x) \in (0,x) \subseteq (0,1)$. By induction $x_n \in (0,1)$ for all n. *Monotonicity:* by a), $x_{n+1} = f(x_n) < x_n$, so (x_n) is strictly decreasing. *Conclusion:* (x_n) is decreasing and bounded below by 0, hence convergent by the monotone convergence theorem; let $L := \lim x_n \ge 0$. Since f is continuous, $L = \lim x_{n+1} = \lim f(x_n) = f(L)$, so L is a fixed point, and by a) $L = 0$. Thus $x_n \to 0$ for every $x_0 \in (0,1)$. $\square$

EXAMPLE: model answers — box questions (3 P each, only the result counts)

Limit. $\lim_{x\to 0} \frac{\sin x - x + x^3/6}{x^5}$: insert $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \ldots \Rightarrow$ numerator $= \frac{x^5}{120} + O(x^7) \Rightarrow$ result $\frac{1}{120}$.
ODE. $y'' - y = e^x$: $\lambda = \pm 1$; $\mu = 1$ is a simple root $\Rightarrow$ resonance $\Rightarrow y_p = Axe^x$, $y_p'' - y_p = 2Ae^x \Rightarrow A = \frac{1}{2} \Rightarrow y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x$.
Integral. $\int \frac{dx}{1+\cos x}$: $1 + \cos x = 2\cos^2 \frac{x}{2}$, substitute $u = \frac{x}{2}$, $dx = 2\,du \Rightarrow \int \frac{du}{\cos^2 u} = \tan u \Rightarrow \tan \frac{x}{2} + C$.
Radius. $\sum \frac{n^n (x-2)^n}{(2n)!}$: coefficient ratio $\to 0 \Rightarrow R = \infty \Rightarrow$ converges for all $x \in \mathbb{R}$.
Improper. $\int_1^\infty \frac{\cos t}{t}\,dt$ conv. (parts); $\int_1^\infty \frac{\cos^2 t}{t}\,dt$ div. ($\cos^2 t = \frac{1+\cos 2t}{2}$); $\int_0^\infty \frac{\sin t}{t^3}\,dt$ div. (at $t = 0$).
Derivative table trap. $\int \frac{dx}{2\cos^2(x/2)} = \tan \frac{x}{2} + C$, *not* $2 \tan \frac{x}{2}$.